Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

DAIRY FACILITY INFORMATION

A. NAME OF DAIRY OR BUSINESS OPERATING THE DAIRY: Vaz Dairy

Physical address of dairy:

Number and Street	City	County	Zip Code
21876 Sunset RD	Los Banos	Merced	93635

Street and nearest cross street (if no address):

Date facility was originally placed in operation: 01/01/1920

Regional Water Quality Control Board Basin Plan designation: San Joaquin River Basin

County Assessor Parcel Number(s) for dairy facility:

X081-X150-X031-XXXX X083-X180-X013-XXXX

B. OPERATORS

Vaz, Evaristo

Operator name: Vaz, Evaristo

Telephone no.: (209) 826-0109 (Landline) (Cellular)

Mailing Address Number and Street	City	State	Zip Code
21876 Sunset AVE	Los Banos	CA	93635

This operator is responsible for paying permit fees.

Vaz, Michael

Operator name: Vaz, Michael

Telephone no.: (209) 509-9293 (Cellular)

Mailing Address Number and Street	City	State	Zip Code
21876 Sunset AVE	Los Banos	CA	93635

C. OWNERS

Vaz, Evaristo

Legal owner name: Vaz, Evaristo

Telephone no.: (209) 826-0109

Landline	Cellular

Mailing Address Number and Street	City	State	Zip Code
21876 Sunset AVE	Los Banos	CA	93635

This owner is responsible for paying permit fees.

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AVAILABLE NUTRIENTS

A. HERD INFORMATION

	Milk Cows	Dry Cows	Bred Heifers (15-24 mo.)	Heifers (7-14 mo. to breeding)	Calves (4-6 mo.)	Calves (0-3 mo.)
Number open confinement	0	0	250	500	150	0
Number under roof	885	150	0	0	0	0
Maximum number	885	150	300	600	250	0
Average number	885	150	150	0	0	0
Avg live weight (lbs)	1,400	1,450	900	650		

Predominant milk cow breed: Holstein

Average milk production: 65 *pounds per cow per day*

B. MANURE GENERATED

Total manure excreted by the herd: 25,889.42 *tons per reporting period*

Total nitrogen from manure: 331,631.40 *lbs per reporting period*

After ammonia losses (30% loss applied): 232,141.98 *lbs per reporting period*

Total phosphorus from manure: 55,022.39 *lbs per reporting period*

Total potassium from manure: 166,578.44 *lbs per reporting period*

Total salt from manure: 451,194.75 *lbs per reporting period*

C. PROCESS WASTEWATER GENERATED

Process wastewater generated: 28,980,000 *gallons*

Total nitrogen generated: 69,956.30 *lbs*

Total phosphorus generated: 25,781.49 *lbs*

Total potassium generated: 136,522.36 *lbs*

Total salt generated: 732,043.43 *lbs*

	28,980,000	*gallons applied*
+	0	*gallons exported*
-	0	*gallons imported*
=	28,980,000	*gallons generated*

D. FRESH WATER SOURCES

Source Description	Type
CCID	Ground water
Charleston New Well	Ground water
Ramos New Well	Ground water
SF IW	Ground water

E. SUBSURFACE (TILE) DRAINAGE SOURCES

No subsurface (tile) drainage sources entered.

F. NUTRIENT IMPORTS

No dry manure nutrient imports entered.

No process wastewater nutrient imports entered.

Date	Material type / Description	Quantity	Reporting basis	Moisture (%)	N (%)	P (%)	K (%)	Salt (%)
04/01/2023	Solid commercial fertilizer *Can 17*	53.00 *ton*	Dry-weight	0.1	17.000000	0.000000	0.000000	0.000000
06/06/2023	Solid commercial fertilizer *UN 32*	25.00 *ton*	Dry-weight	0.1	32.000000	0.000000	0.000000	0.000000

Material type	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Commercial fertilizer / Other	33,985.98	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	0.00	0.00	0.00	0.00
Total imports for all materials	33,985.98	0.00	0.00	0.00

G. NUTRIENT EXPORTS

No solid nutrient exports entered.

No liquid nutrient exports entered.

APPLICATION AREA

A. LIST OF LAND APPLICATION AREAS

Field name	Controlled acres	Cropable acres	Total harvests	Type of waste applied	Parcel number
Charleston East	75	75	2	manure	X088-X082-X010-XXXX
Charleston West	75	75	2	manure	X088-X082-X009-XXXX
Dairy 1	28	28	2	process wastewater	X083-X180-X013-XXXX
Dairy 2	27	27	2	process wastewater	X083-X180-X013-XXXX
Dairy 3	39	39	2	process wastewater	X083-X180-X013-XXXX
Field EV1-3	102	102	1	none	X081-X150-X030-XXXX X081-X150-X031-XXXX X081-X150-X033-XXXX
Field EV4	36	36	2	manure	X081-X150-X032-XXXX
Field EV5	20	20	2	manure	X081-X150-X023-XXXX
Field EV6	24	24	1	manure	X081-X150-X003-XXXX
Field P1-2	67	67	1	none	X084-X010-X080-XXXX
Field P3-4	145	145	2	manure	X084-X010-X080-XXXX
Field R-1	99	99	2	both	X083-X220-X002-XXXX
Field R-2	18	18	2	both	X083-X220-X002-XXXX
Hansen Almonds	87	87	1	none	X078-X280-X016-XXXX
Hansen Open	53	53	2	manure	X078-X280-X019-XXXX
Ingomar Grade	105	92	1	none	X081-X050-X007-XXXX X081-X070-X010-XXXX X081-X080-X001-XXXX
Totals for areas that were used for application	639	639	23		
Totals for areas that were not used for application	361	348	4		
Land application area totals	1,000	987	27		

B. CROPS AND HARVESTS

Charleston East

Field name: Charleston East

Charleston East

11/07/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 75 Plant date: 11/07/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/26/2023	1,200.00 *ton*	As-is		70.0	5,360.00	880.00	6,820.00		11.10

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	14.00	140.00	22.40	116.20	0.00
Total actual harvest content	16.00	171.52	28.16	218.24	1,065.60

06/17/2023: Corn, silage

Crop: Corn, silage Acres planted: 75 Plant date: 06/17/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/07/2023	1,735.00 *ton*	As-is		65.6	4,130.00	600.00	3,680.00		6.34

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	26.00	208.00	39.00	171.60	0.00
Total actual harvest content	23.13	191.08	27.76	170.26	1,009.06

Charleston West

Field name: Charleston West

11/07/2022: Wheat Hay

Crop: Wheat Hay Acres planted: 75 Plant date: 11/07/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
06/06/2023	382.00 *ton*	As-is		6.4	10,100.00	2,090.00	11,200.00		10.00

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	4.00	132.00	20.40	99.60	0.00
Total actual harvest content	5.09	102.89	21.29	114.09	953.47

Charleston West

06/26/2023: Corn, silage

Crop: Corn, silage Acres planted: 75 Plant date: 06/26/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/08/2023	1,801.00 *ton*	As-is		65.6	4,130.00	600.00	3,680.00		6.34

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	26.00	208.00	39.00	171.60	0.00
Total actual harvest content	24.01	198.35	28.82	176.74	1,047.44

Dairy 1

Field name: Dairy 1

11/02/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 28 Plant date: 11/02/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/26/2023	480.00 *ton*	As-is		70.0	6,610.00	1,270.00	13,500.00		17.40

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	17.50	175.00	28.00	145.25	0.00
Total actual harvest content	17.14	226.63	43.54	462.86	1,789.71

06/16/2023: Corn, silage

Crop: Corn, silage Acres planted: 28 Plant date: 06/16/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/03/2023	790.00 *ton*	As-is		65.4	4,120.00	600.00	3,520.00		5.31

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	28.00	224.00	42.00	184.80	0.00
Total actual harvest content	28.21	232.49	33.86	198.63	1,036.74

Dairy 2

Field name: Dairy 2

11/02/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 27 Plant date: 11/02/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/26/2023	444.00 *ton*	As-is		70.0	7,190.00	1,350.00	13,800.00		14.00

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	17.50	175.00	28.00	145.25	0.00
Total actual harvest content	16.44	236.47	44.40	453.87	1,381.33

06/07/2023: Corn, silage

Crop: Corn, silage Acres planted: 27 Plant date: 06/07/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/03/2023	750.00 *ton*	As-is		65.4	4,120.00	600.00	3,520.00		5.31

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	28.00	224.00	42.00	184.80	0.00
Total actual harvest content	27.78	228.89	33.33	195.56	1,020.70

Dairy 3

Field name: Dairy 3

11/02/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 39 Plant date: 11/02/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/27/2023	720.00 *ton*	As-is		70.0	7,020.00	1,500.00	15,100.00		14.30

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	17.50	175.00	28.00	145.25	0.00
Total actual harvest content	18.46	259.20	55.38	557.54	1,584.00

Dairy 3

06/01/2023: Corn, silage

Crop: Corn, silage Acres planted: 39 Plant date: 06/01/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/03/2023	1,052.00 *ton*	As-is		65.4	4,120.00	600.00	3,520.00		5.31

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	28.00	224.00	42.00	184.80	0.00
Total actual harvest content	26.97	222.27	32.37	189.90	991.18

Field EV1-3

Field name: Field EV1-3

01/01/2023: Alfalfa, hay

Crop: Alfalfa, hay Acres planted: 102 Plant date: 01/01/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
11/01/2023	832.00 *ton*	As-is		7.4	27,500.00	2,600.00	19,700.00		10.00

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	6.00	360.00	32.40	252.00	0.00
Total actual harvest content	8.16	448.63	42.42	321.38	1,510.65

Field EV4

Field name: Field EV4

Field EV4

10/26/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 36 Plant date: 10/26/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/29/2023	435.00 *ton*	As-is		70.0	3,340.00	900.00	6,950.00		10.30

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	14.00	140.00	22.40	116.20	0.00
Total actual harvest content	12.08	80.72	21.75	167.96	746.75

06/08/2023: Corn, silage

Crop: Corn, silage Acres planted: 36 Plant date: 06/08/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/06/2023	654.00 *ton*	As-is		57.6	4,620.00	140.00	590.00		7.23

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	20.00	160.00	30.00	132.00	0.00
Total actual harvest content	18.17	167.86	5.09	21.44	1,113.81

Field EV5

Field name: Field EV5

10/26/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 20 Plant date: 10/26/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/29/2023	230.00 *ton*	As-is		70.0	3,340.00	900.00	6,950.00		10.30

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	14.00	140.00	22.40	116.20	0.00
Total actual harvest content	11.50	76.82	20.70	159.85	710.70

Field EV5

06/08/2023: Corn, silage

Crop: Corn, silage Acres planted: 20 Plant date: 06/08/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/06/2023	340.00 *ton*	As-is		57.6	4,620.00	140.00	590.00		7.23

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	20.00	160.00	30.00	132.00	0.00
Total actual harvest content	17.00	157.08	4.76	20.06	1,042.28

Field EV6

Field name: Field EV6

10/26/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 24 Plant date: 10/26/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/29/2023	260.00 *ton*	As-is		70.0	3,340.00	900.00	6,950.00		10.30

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	14.00	140.00	22.40	116.20	0.00
Total actual harvest content	10.83	72.37	19.50	150.58	669.50

Field P1-2

Field name: Field P1-2

Field P1-2

11/09/2022: Wheat Hay

Crop: Wheat Hay Acres planted: 67 Plant date: 11/09/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
06/06/2023	261.00 *ton*	As-is		6.4	10,100.00	2,090.00	11,200.00		10.00

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	4.00	132.00	20.40	99.60	0.00
Total actual harvest content	3.90	78.69	16.28	87.26	729.24

Field P3-4

Field name: Field P3-4

11/10/2022: Wheat Hay

Crop: Wheat Hay Acres planted: 75 Plant date: 11/10/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
06/06/2023	403.00 *ton*	As-is		6.4	10,100.00	2,090.00	11,200.00		10.00

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	3.00	99.00	15.30	74.70	0.00
Total actual harvest content	5.37	108.54	22.46	120.36	1,005.89

07/02/2023: Corn, silage

Crop: Corn, silage Acres planted: 75 Plant date: 07/02/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/20/2023	1,026.00 *ton*	As-is		66.0	3,230.00	840.00	3,040.00		5.60

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	20.00	160.00	30.00	132.00	0.00
Total actual harvest content	13.68	88.37	22.98	83.17	520.93

Field R-1

Field name: Field R-1

10/28/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 99 Plant date: 10/28/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/28/2023	1,560.00 *ton*	As-is		70.0	6,300.00	1,300.00	13,700.00		13.20

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	14.00	140.00	22.40	116.20	0.00
Total actual harvest content	15.76	198.55	40.97	431.76	1,248.00

06/11/2023: Corn, silage

Crop: Corn, silage Acres planted: 99 Plant date: 06/11/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/04/2023	2,526.00 *ton*	As-is		65.8	4,070.00	550.00	3,430.00		5.63

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	24.00	192.00	36.00	158.40	0.00
Total actual harvest content	25.52	207.69	28.07	175.03	982.57

Field R-2

Field name: Field R-2

10/28/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 18 Plant date: 10/28/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/29/2023	302.00 *ton*	As-is		70.0	5,970.00	1,360.00	14,100.00		13.70

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	14.00	140.00	22.40	116.20	0.00
Total actual harvest content	16.78	200.33	45.64	473.13	1,379.13

Field R-2

06/18/2023: Corn, silage

Crop: Corn, silage Acres planted: 18 Plant date: 06/18/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/04/2023	500.00 *ton*	As-is		65.8	4,070.00	550.00	3,430.00		5.63

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	24.00	192.00	36.00	158.40	0.00
Total actual harvest content	27.78	226.11	30.56	190.56	1,069.70

Hansen Almonds

Field name: Hansen Almonds

01/01/2023: Almond Field Run

Crop: Almond Field Run Acres planted: 87 Plant date: 01/01/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
11/01/2023	113.00 *ton*	As-is		10.0	65,000.00	10,000.00	50,000.00		0.00

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	4.00	172.00	29.20	106.00	0.00
Total actual harvest content	1.30	168.85	25.98	129.89	0.00

Hansen Open

Field name: Hansen Open

Hansen Open

10/30/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 53 Plant date: 10/30/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/29/2023	832.00 *ton*	As-is		70.0	3,400.00	860.00	6,550.00		10.60

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	14.00	140.00	22.40	116.20	0.00
Total actual harvest content	15.70	106.75	27.00	205.65	998.40

06/06/2023: Corn, silage

Crop: Corn, silage Acres planted: 53 Plant date: 06/06/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/05/2023	907.00 *ton*	As-is		59.2	4,370.00	750.00	4,020.00		5.07

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	20.00	200.00	32.00	166.00	0.00
Total actual harvest content	17.11	149.57	25.67	137.59	707.99

Ingomar Grade

Field name: Ingomar Grade

01/01/2023: Almond Field Run

Crop: Almond Field Run Acres planted: 92 Plant date: 01/01/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
11/01/2023	40.00 *ton*	As-is		10.0	65,000.00	10,000.00	50,000.00		0.00

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	4.00	172.00	29.20	106.00	0.00
Total actual harvest content	0.43	56.52	8.70	43.48	0.00

NUTRIENT BUDGET

A. LAND APPLICATIONS

Charleston East - 11/07/2022: Oats, silage-soft dough

Field name: Charleston East

Crop: Oats, silage-soft dough — Plant date: 11/07/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/20/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Corral Solids	Corral solids	115.40	35.74	276.74	2,502.40	400.00 *ton*
Application event totals		115.40	35.74	276.74	2,502.40	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/06/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Charleston New Well	Ground water	28.80	0.00	0.00	0.00	12,096,000.00 *gal*
Application event totals		28.80	0.00	0.00	0.00	

Charleston East - 06/17/2023: Corn, silage

Field name: Charleston East

Crop: Corn, silage — Plant date: 06/17/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Charleston New Well	Ground water	41.15	0.00	0.00	0.00	17,280,000.00 *gal*
Application event totals		41.15	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/17/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
UN 32	Solid commercial fertilizer	80.00	0.00	0.00	0.00	
Application event totals		80.00	0.00	0.00	0.00	

Charleston East - 06/17/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Charleston New Well	Ground water	32.92	0.00	0.00	0.00	13,824,000.00 *gal*
Application event totals		32.92	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Charleston New Well	Ground water	24.69	0.00	0.00	0.00	10,368,000.00 *gal*
Application event totals		24.69	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Charleston New Well	Ground water	26.74	0.00	0.00	0.00	11,232,000.00 *gal*
Application event totals		26.74	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/27/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Charleston New Well	Ground water	24.69	0.00	0.00	0.00	10,368,000.00 *gal*
Application event totals		24.69	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/13/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Charleston New Well	Ground water	24.69	0.00	0.00	0.00	10,368,000.00 *gal*
Application event totals		24.69	0.00	0.00	0.00	

Charleston West - 11/07/2022: Wheat Hay

Field name: Charleston West

Crop: Wheat Hay

Plant date: 11/07/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following

Charleston West - 11/07/2022: Wheat Hay

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/20/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Corral Solids	Corral solids	115.40	35.74	276.74	2,502.40	400.00 *ton*
Application event totals		115.40	35.74	276.74	2,502.40	

03/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Charleston New Well	Ground water	28.80	0.00	0.00	0.00	12,096,000.00 *gal*
Application event totals		28.80	0.00	0.00	0.00	

Charleston West - 06/26/2023: Corn, silage

Field name: Charleston West

Crop: Corn, silage

Plant date: 06/26/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/09/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Charleston New Well	Ground water	49.37	0.00	0.00	0.00	20,736,000.00 *gal*
Application event totals		49.37	0.00	0.00	0.00	

06/26/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
UN 32	Solid commercial fertilizer	80.00	0.00	0.00	0.00	
Application event totals		80.00	0.00	0.00	0.00	

07/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Charleston New Well	Ground water	28.80	0.00	0.00	0.00	12,096,000.00 *gal*
Application event totals		28.80	0.00	0.00	0.00	

Charleston West - 06/26/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Charleston New Well	Ground water	24.69	0.00	0.00	0.00	10,368,000.00 *gal*
Application event totals		24.69	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Charleston New Well	Ground water	24.69	0.00	0.00	0.00	10,368,000.00 *gal*
Application event totals		24.69	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Charleston New Well	Ground water	24.69	0.00	0.00	0.00	10,368,000.00 *gal*
Application event totals		24.69	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Charleston New Well	Ground water	28.80	0.00	0.00	0.00	12,096,000.00 *gal*
Application event totals		28.80	0.00	0.00	0.00	

Dairy 1 - 11/02/2022: Oats, silage-soft dough

Field name: Dairy 1

Crop: Oats, silage-soft dough

Plant date: 11/02/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/10/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	119.10	24.93	179.18	1,416.27	1,200,000.00 *gal*
CCID	Ground water	0.00	0.00	0.00	149.67	3,240,000.00 *gal*
Application event totals		119.10	24.93	179.18	1,565.94	

Dairy 1 - 11/02/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	159.65	19.54	220.49	1,744.58	1,080,000.00 *gal*
CCID	Ground water	0.00	0.00	0.00	134.71	2,916,000.00 *gal*
Application event totals		159.65	19.54	220.49	1,879.29	

Dairy 1 - 06/16/2023: Corn, silage

Field name: Dairy 1

Crop: Corn, silage

Plant date: 06/16/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	93.56	51.07	291.84	274.67	1,440,000.00 *gal*
CCID	Ground water	0.00	0.00	0.00	269.41	5,832,000.00 *gal*
Application event totals		93.56	51.07	291.84	544.08	

07/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID	Ground water	0.00	0.00	0.00	224.51	4,860,000.00 *gal*
Application event totals		0.00	0.00	0.00	224.51	

07/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	70.17	38.30	218.88	206.00	1,080,000.00 *gal*
CCID	Ground water	0.00	0.00	0.00	202.06	4,374,000.00 *gal*
Application event totals		70.17	38.30	218.88	408.06	

Dairy 1 - 06/16/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID	Ground water	0.00	0.00	0.00	179.61	3,888,000.00 *gal*
Application event totals		0.00	0.00	0.00	179.61	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	46.78	25.54	145.92	137.33	720,000.00 *gal*
CCID	Ground water	0.00	0.00	0.00	149.67	3,240,000.00 *gal*
Application event totals		46.78	25.54	145.92	287.01	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/04/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	66.74	11.69	79.40	613.72	720,000.00 *gal*
CCID	Ground water	0.00	0.00	0.00	157.16	3,402,000.00 *gal*
Application event totals		66.74	11.69	79.40	770.87	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID	Ground water	0.00	0.00	0.00	179.61	3,888,000.00 *gal*
Application event totals		0.00	0.00	0.00	179.61	

Dairy 2 - 11/02/2022: Oats, silage-soft dough

Field name: Dairy 2

Crop: Oats, silage-soft dough

Plant date: 11/02/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following

Dairy 2 - 11/02/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/12/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	129.68	27.14	195.11	1,542.16	1,260,000.00 *gal*
CCID	Ground water	0.00	0.00	0.00	162.98	3,402,000.00 *gal*
Application event totals		129.68	27.14	195.11	1,705.13	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/04/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	165.56	20.26	228.65	1,809.20	1,080,000.00 *gal*
CCID	Ground water	0.00	0.00	0.00	139.70	2,916,000.00 *gal*
Application event totals		165.56	20.26	228.65	1,948.89	

Dairy 2 - 06/07/2023: Corn, silage

Field name: Dairy 2

Crop: Corn, silage

Plant date: 06/07/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	97.02	52.96	302.65	284.84	1,440,000.00 *gal*
CCID	Ground water	0.00	0.00	0.00	232.83	4,860,000.00 *gal*
Application event totals		97.02	52.96	302.65	517.67	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID	Ground water	0.00	0.00	0.00	209.54	4,374,000.00 *gal*
Application event totals		0.00	0.00	0.00	209.54	

Dairy 2 - 06/07/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	72.77	39.72	226.98	213.63	1,080,000.00 *gal*
CCID	Ground water	0.00	0.00	0.00	190.14	3,969,000.00 *gal*
Application event totals		72.77	39.72	226.98	403.77	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/31/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID	Ground water	0.00	0.00	0.00	186.26	3,888,000.00 *gal*
Application event totals		0.00	0.00	0.00	186.26	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	72.77	39.72	226.98	213.63	1,080,000.00 *gal*
CCID	Ground water	0.00	0.00	0.00	139.70	2,916,000.00 *gal*
Application event totals		72.77	39.72	226.98	353.33	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	69.21	12.13	82.34	636.45	720,000.00 *gal*
CCID	Ground water	0.00	0.00	0.00	186.26	3,888,000.00 *gal*
Application event totals		69.21	12.13	82.34	822.71	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID	Ground water	0.00	0.00	0.00	162.98	3,402,000.00 *gal*
Application event totals		0.00	0.00	0.00	162.98	

Dairy 3 - 11/02/2022: Oats, silage-soft dough

Field name: Dairy 3

Crop: Oats, silage-soft dough

Plant date: 11/02/2022

Dairy 3 - 11/02/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/10/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	163.15	11.79	295.34	2,560.50	2,160,000.00 *gal*
CCID	Ground water	0.00	0.00	0.00	193.42	5,832,000.00 *gal*
Application event totals		163.15	11.79	295.34	2,753.93	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	92.82	15.44	124.40	1,348.04	1,800,000.00 *gal*
CCID	Ground water	0.00	0.00	0.00	161.19	4,860,000.00 *gal*
Application event totals		92.82	15.44	124.40	1,509.23	

Dairy 3 - 06/01/2023: Corn, silage

Field name: Dairy 3

Crop: Corn, silage

Plant date: 06/01/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	50.38	27.50	157.14	147.90	1,080,000.00 *gal*
CCID	Ground water	0.00	0.00	0.00	193.42	5,832,000.00 *gal*
Application event totals		50.38	27.50	157.14	341.32	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID	Ground water	0.00	0.00	0.00	225.66	6,804,000.00 *gal*
Application event totals		0.00	0.00	0.00	225.66	

Dairy 3 - 06/01/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/13/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	67.17	36.67	209.52	197.20	1,440,000.00 *gal*
CCID	Ground water	0.00	0.00	0.00	182.68	5,508,000.00 *gal*
Application event totals		67.17	36.67	209.52	379.88	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	50.38	27.50	157.14	147.90	1,080,000.00 *gal*
CCID	Ground water	0.00	0.00	0.00	145.07	4,374,000.00 *gal*
Application event totals		50.38	27.50	157.14	292.97	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID	Ground water	0.00	0.00	0.00	166.56	5,022,000.00 *gal*
Application event totals		0.00	0.00	0.00	166.56	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	71.87	12.59	85.50	660.92	1,080,000.00 *gal*
CCID	Ground water	0.00	0.00	0.00	171.93	5,184,000.00 *gal*
Application event totals		71.87	12.59	85.50	832.86	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID	Ground water	0.00	0.00	0.00	177.31	5,346,000.00 *gal*
Application event totals		0.00	0.00	0.00	177.31	

Field EV1-3 - 01/01/2023: Alfalfa, hay

Field name: Field EV1-3

Crop: Alfalfa, hay

Plant date: 01/01/2023

Field EV1-3 - 01/01/2023: Alfalfa, hay

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
02/27/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID	Ground water	0.00	0.00	0.00	301.30	23,760,000.00 *gal*
Application event totals		0.00	0.00	0.00	301.30	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID	Ground water	0.00	0.00	0.00	295.83	23,328,000.00 *gal*
Application event totals		0.00	0.00	0.00	295.83	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID	Ground water	0.00	0.00	0.00	230.09	18,144,000.00 *gal*
Application event totals		0.00	0.00	0.00	230.09	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID	Ground water	0.00	0.00	0.00	230.09	18,144,000.00 *gal*
Application event totals		0.00	0.00	0.00	230.09	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID	Ground water	0.00	0.00	0.00	262.96	20,736,000.00 *gal*
Application event totals		0.00	0.00	0.00	262.96	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID	Ground water	0.00	0.00	0.00	230.09	18,144,000.00 *gal*
Application event totals		0.00	0.00	0.00	230.09	

Field EV1-3 - 01/01/2023: Alfalfa, hay

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID	Ground water	0.00	0.00	0.00	230.09	18,144,000.00 *gal*
Application event totals		0.00	0.00	0.00	230.09	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID	Ground water	0.00	0.00	0.00	230.09	18,144,000.00 *gal*
Application event totals		0.00	0.00	0.00	230.09	

Field EV4 - 10/26/2022: Oats, silage-soft dough

Field name: Field EV4

Crop: Oats, silage-soft dough Plant date: 10/26/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/10/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Corral solids	100.75	54.56	235.60	5,859.00	250.00 *ton*
Application event totals		100.75	54.56	235.60	5,859.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/04/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID	Ground water	0.00	0.00	0.00	139.70	3,888,000.00 *gal*
Application event totals		0.00	0.00	0.00	139.70	

Field EV4 - 06/08/2023: Corn, silage

Field name: Field EV4

Crop: Corn, silage Plant date: 06/08/2023

Field EV4 - 06/08/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/03/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Separator Solids	Separator solids	172.87	44.05	173.85	0.00	600.00 *ton*
Application event totals		172.87	44.05	173.85	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID	Ground water	0.00	0.00	0.00	209.54	5,832,000.00 *gal*
Application event totals		0.00	0.00	0.00	209.54	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID	Ground water	0.00	0.00	0.00	279.39	7,776,000.00 *gal*
Application event totals		0.00	0.00	0.00	279.39	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID	Ground water	0.00	0.00	0.00	209.54	5,832,000.00 *gal*
Application event totals		0.00	0.00	0.00	209.54	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID	Ground water	0.00	0.00	0.00	209.54	5,832,000.00 *gal*
Application event totals		0.00	0.00	0.00	209.54	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID	Ground water	0.00	0.00	0.00	349.24	9,720,000.00 *gal*
Application event totals		0.00	0.00	0.00	349.24	

Field EV4 - 06/08/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID	Ground water	0.00	0.00	0.00	139.70	3,888,000.00 *gal*
Application event totals		0.00	0.00	0.00	139.70	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID	Ground water	0.00	0.00	0.00	139.70	3,888,000.00 *gal*
Application event totals		0.00	0.00	0.00	139.70	

Field EV5 - 10/26/2022: Oats, silage-soft dough

Field name: Field EV5

Crop: Oats, silage-soft dough Plant date: 10/26/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/04/2021	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Corral solids	90.68	49.10	212.04	5,273.10	125.00 *ton*
Application event totals		90.68	49.10	212.04	5,273.10	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID	Ground water	0.00	0.00	0.00	377.18	5,832,000.00 *gal*
Application event totals		0.00	0.00	0.00	377.18	

Field EV5 - 06/08/2023: Corn, silage

Field name: Field EV5

Crop: Corn, silage Plant date: 06/08/2023

Field EV5 - 06/08/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/03/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Separator Solids	Separator solids	155.58	39.64	156.46	0.00	300.00 *ton*
Application event totals		155.58	39.64	156.46	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID	Ground water	0.00	0.00	0.00	502.90	7,776,000.00 *gal*
Application event totals		0.00	0.00	0.00	502.90	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID	Ground water	0.00	0.00	0.00	377.18	5,832,000.00 *gal*
Application event totals		0.00	0.00	0.00	377.18	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID	Ground water	0.00	0.00	0.00	377.18	5,832,000.00 *gal*
Application event totals		0.00	0.00	0.00	377.18	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID	Ground water	0.00	0.00	0.00	377.18	5,832,000.00 *gal*
Application event totals		0.00	0.00	0.00	377.18	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID	Ground water	0.00	0.00	0.00	377.18	5,832,000.00 *gal*
Application event totals		0.00	0.00	0.00	377.18	

Field EV5 - 06/08/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID	Ground water	0.00	0.00	0.00	251.45	3,888,000.00 *gal*
Application event totals		0.00	0.00	0.00	251.45	

09/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID	Ground water	0.00	0.00	0.00	251.45	3,888,000.00 *gal*
Application event totals		0.00	0.00	0.00	251.45	

Field EV6 - 10/26/2022: Oats, silage-soft dough

Field name: Field EV6

Crop: Oats, silage-soft dough Plant date: 10/26/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/04/2021	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Corral Solids	Corral solids	90.68	49.10	212.04	5,273.10	150.00 *ton*
Application event totals		90.68	49.10	212.04	5,273.10	

02/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID	Ground water	0.00	0.00	0.00	314.31	5,832,000.00 *gal*
Application event totals		0.00	0.00	0.00	314.31	

Field P1-2 - 11/09/2022: Wheat Hay

Field name: Field P1-2

Crop: Wheat Hay Plant date: 11/09/2022

Field P1-2 - 11/09/2022: Wheat Hay

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
02/27/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
SF IW	Ground water	8.76	0.00	0.00	0.00	11,664,000.00 *gal*
Application event totals		8.76	0.00	0.00	0.00	

03/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
SF IW	Ground water	7.30	0.00	0.00	0.00	9,720,000.00 *gal*
Application event totals		7.30	0.00	0.00	0.00	

Field P3-4 - 11/10/2022: Wheat Hay

Field name: Field P3-4

Crop: Wheat Hay

Plant date: 11/10/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/17/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Corral Solids	Corral solids	129.83	40.21	311.33	2,815.20	450.00 *ton*
Application event totals		129.83	40.21	311.33	2,815.20	

03/04/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
SF IW	Ground water	8.70	0.00	0.00	0.00	12,960,000.00 *gal*
Application event totals		8.70	0.00	0.00	0.00	

04/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
SF IW	Ground water	7.83	0.00	0.00	0.00	11,664,000.00 *gal*
Application event totals		7.83	0.00	0.00	0.00	

Field P3-4 - 07/02/2023: Corn, silage

Field name: Field P3-4

Crop: Corn, silage

Plant date: 07/02/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
SF IW	Ground water	9.13	0.00	0.00	0.00	13,608,000.00 *gal*
Application event totals		9.13	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/27/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
SF IW	Ground water	7.83	0.00	0.00	0.00	11,664,000.00 *gal*
Application event totals		7.83	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
SF IW	Ground water	7.17	0.00	0.00	0.00	10,692,000.00 *gal*
Application event totals		7.17	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
SF IW	Ground water	6.74	0.00	0.00	0.00	10,044,000.00 *gal*
Application event totals		6.74	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/09/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
SF IW	Ground water	7.17	0.00	0.00	0.00	10,692,000.00 *gal*
Application event totals		7.17	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
SF IW	Ground water	7.83	0.00	0.00	0.00	11,664,000.00 *gal*
Application event totals		7.83	0.00	0.00	0.00	

Field P3-4 - 07/02/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/06/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
SF IW	Ground water	6.52	0.00	0.00	0.00	9,720,000.00 *gal*
Application event totals		6.52	0.00	0.00	0.00	

Field R-1 - 10/28/2022: Oats, silage-soft dough

Field name: Field R-1

Crop: Oats, silage-soft dough

Plant date: 10/28/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/09/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	99.98	7.22	180.98	1,569.06	3,360,000.00 *gal*
CCID	Ground water	35.07	0.00	0.00	1,600.78	12,096,000.00 *gal*
Application event totals		135.05	7.22	180.98	3,169.85	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
02/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	58.51	97.35	78.41	849.67	2,880,000.00 *gal*
Ramos New Well	Ground water	30.06	0.00	0.00	1,372.10	10,368,000.00 *gal*
Application event totals		88.57	97.35	78.41	2,221.77	

Field R-1 - 06/11/2023: Corn, silage

Field name: Field R-1

Crop: Corn, silage

Plant date: 06/11/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following

Field R-1 - 06/11/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Corral Solids	Separator solids	100.10	57.05	68.89	2,890.04	800.00 *ton*
Application event totals		100.10	57.05	68.89	2,890.04	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ramos New Well	Ground water	35.13	0.00	0.00	0.00	15,552,000.00 *gal*
Application event totals		35.13	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ramos New Well	Ground water	26.35	0.00	0.00	0.00	11,664,000.00 *gal*
Application event totals		26.35	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ramos New Well	Ground water	26.35	0.00	0.00	0.00	11,664,000.00 *gal*
Application event totals		26.35	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/04/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ramos New Well	Ground water	29.28	0.00	0.00	0.00	12,960,000.00 *gal*
Application event totals		29.28	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ramos New Well	Ground water	23.42	0.00	0.00	0.00	10,368,000.00 *gal*
Application event totals		23.42	0.00	0.00	0.00	

Field R-1 - 06/11/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ramos New Well	Ground water	26.35	0.00	0.00	0.00	11,664,000.00 *gal*
Application event totals		26.35	0.00	0.00	0.00	

09/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ramos New Well	Ground water	23.42	0.00	0.00	0.00	10,368,000.00 *gal*
Application event totals		23.42	0.00	0.00	0.00	

Field R-2 - 10/28/2022: Oats, silage-soft dough

Field name: Field R-2

Crop: Oats, silage-soft dough Plant date: 10/28/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/06/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	98.19	7.09	177.75	1,541.04	600,000.00 *gal*
Ramos New Well	Ground water	26.84	0.00	0.00	0.00	2,160,000.00 *gal*
Application event totals		125.03	7.09	177.75	1,541.04	

02/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	67.04	11.15	89.85	973.58	600,000.00 *gal*
Ramos New Well	Ground water	26.84	0.00	0.00	0.00	2,160,000.00 *gal*
Application event totals		93.88	11.15	89.85	973.58	

Field R-2 - 06/18/2023: Corn, silage

Field R-2 - 06/18/2023: Corn, silage

Field name: Field R-2

Crop: Corn, silage

Plant date: 06/18/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/03/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Separator Solids	Separator solids	103.23	58.83	71.04	2,980.35	150.00 *ton*
Application event totals		103.23	58.83	71.04	2,980.35	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ramos New Well	Ground water	37.57	0.00	0.00	0.00	3,024,000.00 *gal*
Application event totals		37.57	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ramos New Well	Ground water	32.21	0.00	0.00	0.00	2,592,000.00 *gal*
Application event totals		32.21	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ramos New Well	Ground water	32.21	0.00	0.00	0.00	2,592,000.00 *gal*
Application event totals		32.21	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ramos New Well	Ground water	26.84	0.00	0.00	0.00	2,160,000.00 *gal*
Application event totals		26.84	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ramos New Well	Ground water	26.84	0.00	0.00	0.00	2,160,000.00 *gal*
Application event totals		26.84	0.00	0.00	0.00	

Field R-2 - 06/18/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ramos New Well	Ground water	21.47	0.00	0.00	0.00	1,728,000.00 *gal*
Application event totals		21.47	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ramos New Well	Ground water	26.84	0.00	0.00	0.00	2,160,000.00 *gal*
Application event totals		26.84	0.00	0.00	0.00	

Hansen Almonds - 01/01/2023: Almond Field Run

Field name: Hansen Almonds

Crop: Almond Field Run

Plant date: 01/01/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID	Ground water	0.00	0.00	0.00	192.68	12,960,000.00 *gal*
Application event totals		0.00	0.00	0.00	192.68	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID	Ground water	0.00	0.00	0.00	385.37	25,920,000.00 *gal*
Application event totals		0.00	0.00	0.00	385.37	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/01/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CAN17	Solid commercial fertilizer	51.00	0.00	0.00	0.00	
Application event totals		51.00	0.00	0.00	0.00	

Hansen Almonds - 01/01/2023: Almond Field Run

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID	Ground water	0.00	0.00	0.00	385.37	25,920,000.00 *gal*
Application event totals		0.00	0.00	0.00	385.37	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/01/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CAN17	Solid commercial fertilizer	51.00	0.00	0.00	0.00	
Application event totals		51.00	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID	Ground water	0.00	0.00	0.00	385.37	25,920,000.00 *gal*
Application event totals		0.00	0.00	0.00	385.37	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/01/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CAN17	Solid commercial fertilizer	51.00	0.00	0.00	0.00	
Application event totals		51.00	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID	Ground water	0.00	0.00	0.00	385.37	25,920,000.00 *gal*
Application event totals		0.00	0.00	0.00	385.37	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID	Ground water	0.00	0.00	0.00	385.37	25,920,000.00 *gal*
Application event totals		0.00	0.00	0.00	385.37	

Hansen Almonds - 01/01/2023: Almond Field Run

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID	Ground water	0.00	0.00	0.00	192.68	12,960,000.00 *gal*
Application event totals		0.00	0.00	0.00	192.68	

10/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID	Ground water	0.00	0.00	0.00	192.68	12,960,000.00 *gal*
Application event totals		0.00	0.00	0.00	192.68	

Hansen Open - 10/30/2022: Oats, silage-soft dough

Field name: Hansen Open

Crop: Oats, silage-soft dough Plant date: 10/30/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/18/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Corral Solids	Corral solids	142.90	44.25	342.66	3,098.49	350.00 *ton*
Application event totals		142.90	44.25	342.66	3,098.49	

03/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID	Ground water	0.00	0.00	0.00	189.77	7,776,000.00 *gal*
Application event totals		0.00	0.00	0.00	189.77	

Hansen Open - 06/06/2023: Corn, silage

Field name: Hansen Open

Crop: Corn, silage Plant date: 06/06/2023

Hansen Open - 06/06/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID	Ground water	0.00	0.00	0.00	189.77	7,776,000.00 *gal*
Application event totals		0.00	0.00	0.00	189.77	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/06/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
UN 32	Solid commercial fertilizer	80.00	0.00	0.00	0.00	
Application event totals		80.00	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID	Ground water	0.00	0.00	0.00	142.33	5,832,000.00 *gal*
Application event totals		0.00	0.00	0.00	142.33	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID	Ground water	0.00	0.00	0.00	142.33	5,832,000.00 *gal*
Application event totals		0.00	0.00	0.00	142.33	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/31/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID	Ground water	0.00	0.00	0.00	142.33	5,832,000.00 *gal*
Application event totals		0.00	0.00	0.00	142.33	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID	Ground water	0.00	0.00	0.00	142.33	5,832,000.00 *gal*
Application event totals		0.00	0.00	0.00	142.33	

Hansen Open - 06/06/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID	Ground water	0.00	0.00	0.00	94.89	3,888,000.00 *gal*
Application event totals		0.00	0.00	0.00	94.89	

09/13/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID	Ground water	0.00	0.00	0.00	94.89	3,888,000.00 *gal*
Application event totals		0.00	0.00	0.00	94.89	

Ingomar Grade - 01/01/2023: Almond Field Run

Field name: Ingomar Grade

Crop: Almond Field Run

Plant date: 01/01/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/01/2022	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CAN17	Solid commercial fertilizer	25.50	0.00	0.00	0.00	
Application event totals		25.50	0.00	0.00	0.00	

05/01/2022	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CAN17	Solid commercial fertilizer	25.50	0.00	0.00	0.00	
Application event totals		25.50	0.00	0.00	0.00	

03/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID	Ground water	0.00	0.00	0.00	182.21	12,960,000.00 *gal*
Application event totals		0.00	0.00	0.00	182.21	

Ingomar Grade - 01/01/2023: Almond Field Run

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID	Ground water	0.00	0.00	0.00	364.42	25,920,000.00 *gal*
Application event totals		0.00	0.00	0.00	364.42	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID	Ground water	0.00	0.00	0.00	364.42	25,920,000.00 *gal*
Application event totals		0.00	0.00	0.00	364.42	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID	Ground water	0.00	0.00	0.00	364.42	25,920,000.00 *gal*
Application event totals		0.00	0.00	0.00	364.42	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID	Ground water	0.00	0.00	0.00	364.42	25,920,000.00 *gal*
Application event totals		0.00	0.00	0.00	364.42	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID	Ground water	0.00	0.00	0.00	364.42	25,920,000.00 *gal*
Application event totals		0.00	0.00	0.00	364.42	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID	Ground water	0.00	0.00	0.00	364.42	25,920,000.00 *gal*
Application event totals		0.00	0.00	0.00	364.42	

Ingomar Grade - 01/01/2023: Almond Field Run

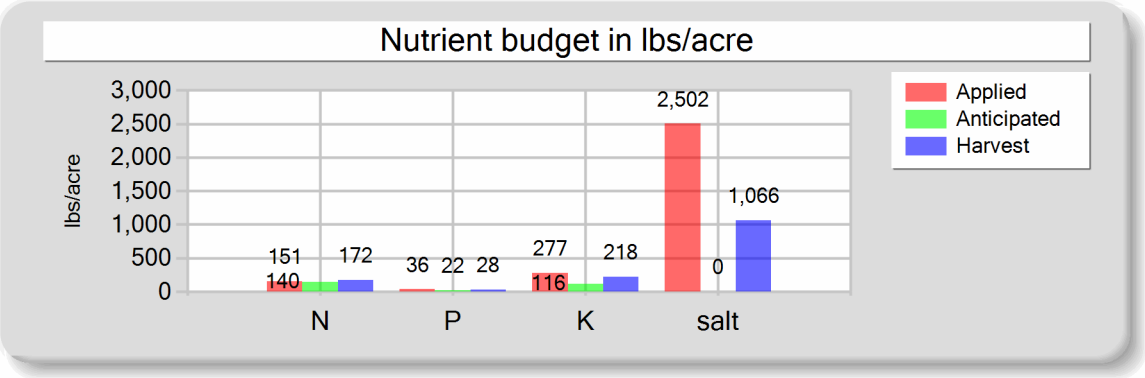
Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID	Ground water	0.00	0.00	0.00	182.21	12,960,000.00 *gal*
Application event totals		0.00	0.00	0.00	182.21	

B. NUTRIENT BUDGET

Charleston East - 11/07/2022: Oats, silage-soft dough

Field name: Charleston East | Crop: Oats, silage-soft dough | Plant date: 11/07/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	115.40	35.74	276.74	2,502.40
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	28.80	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	151.21	35.74	276.74	2,502.40
Anticipated crop nutrient removal	140.00	22.40	116.20	0.00
Actual crop nutrient removal	171.52	28.16	218.24	1,065.60
Nutrient balance	-20.31	7.58	58.50	1,436.80
Applied to removed ratio	0.88	1.27	1.27	2.35

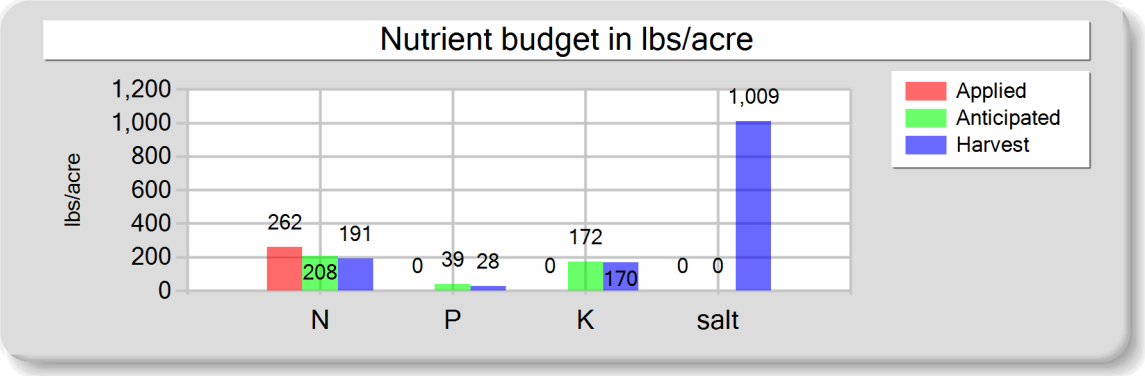
Fresh water applied	
12,096,000.00	*gallons*
445.45	*acre-inches*
5.94	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

Charleston East - 06/17/2023: Corn, silage

Field name: Charleston East Crop: Corn, silage Plant date: 06/17/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	80.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	174.87	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	261.87	0.00	0.00	0.00
Anticipated crop nutrient removal	208.00	39.00	171.60	0.00
Actual crop nutrient removal	191.08	27.76	170.26	1,009.06
Nutrient balance	70.79	-27.76	-170.26	-1,009.06
Applied to removed ratio	1.37	0.00	0.00	0.00

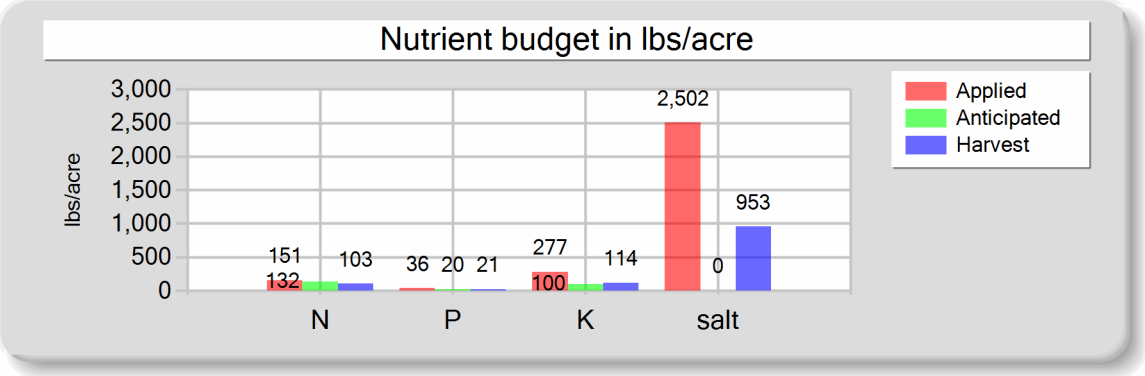
Fresh water applied	
73,440,000.00	*gallons*
2,704.55	*acre-inches*
36.06	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

Charleston West - 11/07/2022: Wheat Hay

Field name: Charleston West Crop: Wheat Hay Plant date: 11/07/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	115.40	35.74	276.74	2,502.40
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	28.80	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	151.21	35.74	276.74	2,502.40
Anticipated crop nutrient removal	132.00	20.40	99.60	0.00
Actual crop nutrient removal	102.89	21.29	114.09	953.47
Nutrient balance	48.32	14.45	162.65	1,548.93
Applied to removed ratio	1.47	1.68	2.43	2.62

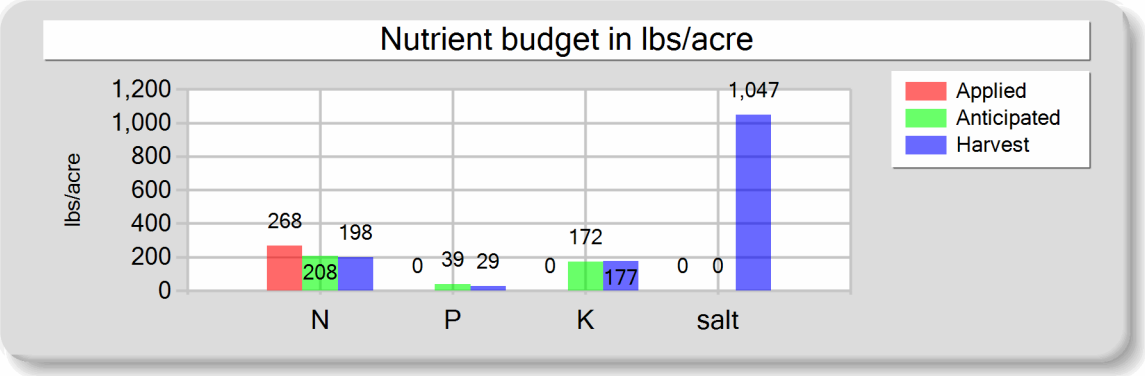
Fresh water applied	
12,096,000.00	*gallons*
445.45	*acre-inches*
5.94	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

Charleston West - 06/26/2023: Corn, silage

Field name: Charleston West Crop: Corn, silage Plant date: 06/26/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	80.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	181.04	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	268.04	0.00	0.00	0.00
Anticipated crop nutrient removal	208.00	39.00	171.60	0.00
Actual crop nutrient removal	198.35	28.82	176.74	1,047.44
Nutrient balance	69.69	-28.82	-176.74	-1,047.44
Applied to removed ratio	1.35	0.00	0.00	0.00

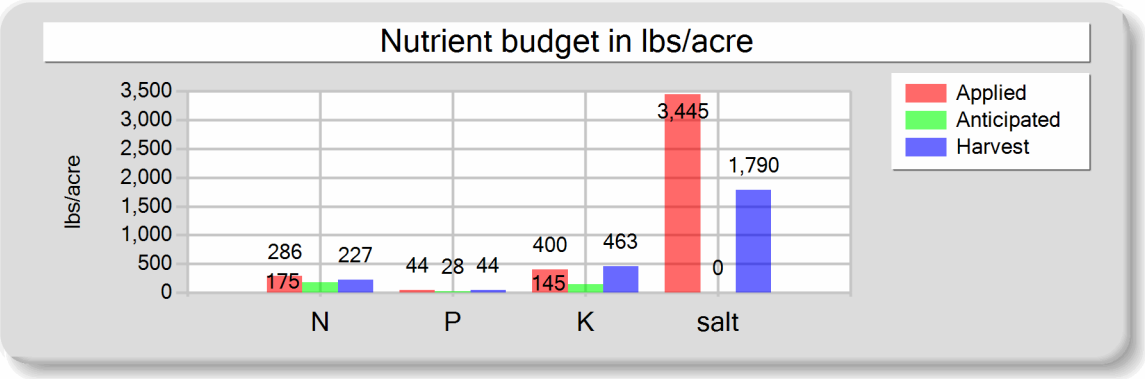
Fresh water applied	
76,032,000.00	*gallons*
2,800.00	*acre-inches*
37.33	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

Dairy 1 - 11/02/2022: Oats, silage-soft dough

Field name: Dairy 1 Crop: Oats, silage-soft dough Plant date: 11/02/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	278.75	44.47	399.67	3,160.85
Fresh water	0.00	0.00	0.00	284.38
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	285.75	44.47	399.67	3,445.23
Anticipated crop nutrient removal	175.00	28.00	145.25	0.00
Actual crop nutrient removal	226.63	43.54	462.86	1,789.71
Nutrient balance	59.12	0.92	-63.19	1,655.51
Applied to removed ratio	1.26	1.02	0.86	1.93

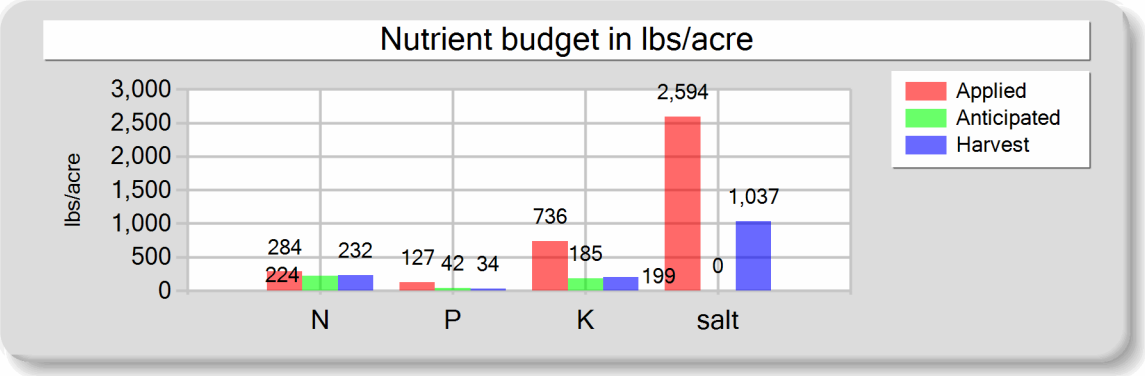
Fresh water applied	
6,156,000.00	*gallons*
226.70	*acre-inches*
8.10	*inches/acre*

Process wastewater applied	
2,280,000.00	*gallons*
83.96	*acre-inches*
3.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

Dairy 1 - 06/16/2023: Corn, silage

Field name: Dairy 1 Crop: Corn, silage Plant date: 06/16/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	277.24	126.61	736.03	1,231.72
Fresh water	0.00	0.00	0.00	1,362.03
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	284.24	126.61	736.03	2,593.75
Anticipated crop nutrient removal	224.00	42.00	184.80	0.00
Actual crop nutrient removal	232.49	33.86	198.63	1,036.74
Nutrient balance	51.76	92.75	537.40	1,557.01
Applied to removed ratio	1.22	3.74	3.71	2.50

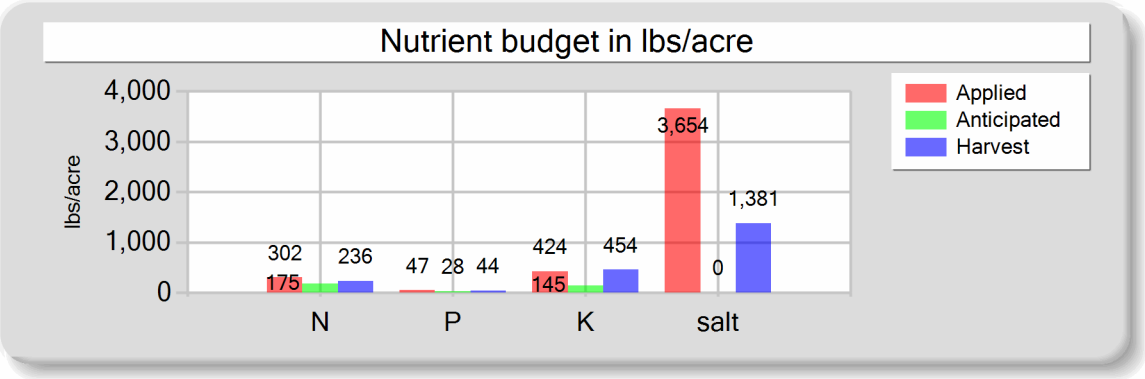
Fresh water applied	
29,484,000.00	*gallons*
1,085.80	*acre-inches*
38.78	*inches/acre*

Process wastewater applied	
3,960,000.00	*gallons*
145.83	*acre-inches*
5.21	*inches/acre*

Total harvests for the crop	
1	*harvests*

Dairy 2 - 11/02/2022: Oats, silage-soft dough

Field name: Dairy 2 Crop: Oats, silage-soft dough Plant date: 11/02/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	295.25	47.41	423.76	3,351.35
Fresh water	0.00	0.00	0.00	302.67
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	302.25	47.41	423.76	3,654.03
Anticipated crop nutrient removal	175.00	28.00	145.25	0.00
Actual crop nutrient removal	236.47	44.40	453.87	1,381.33
Nutrient balance	65.77	3.01	-30.11	2,272.69
Applied to removed ratio	1.28	1.07	0.93	2.65

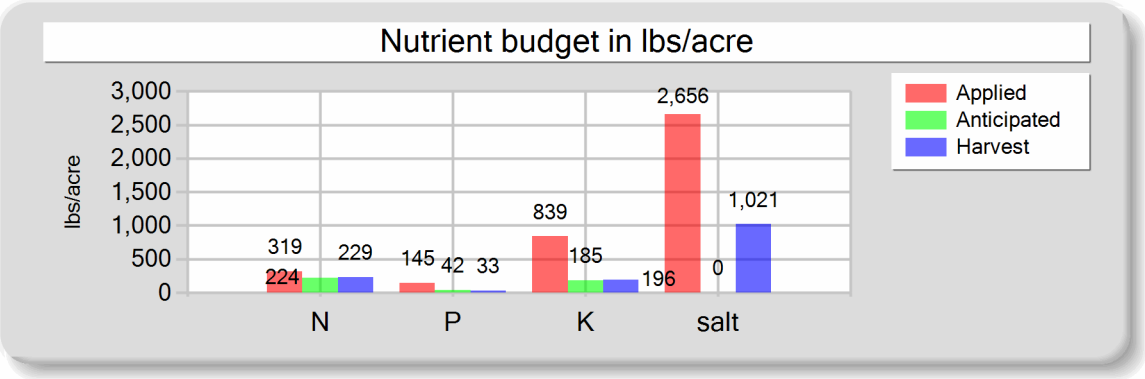
Fresh water applied	
6,318,000.00	*gallons*
232.67	*acre-inches*
8.62	*inches/acre*

Process wastewater applied	
2,340,000.00	*gallons*
86.17	*acre-inches*
3.19	*inches/acre*

Total harvests for the crop	
1	*harvests*

Dairy 2 - 06/07/2023: Corn, silage

Field name: Dairy 2 Crop: Corn, silage Plant date: 06/07/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	311.77	144.54	838.95	1,348.55
Fresh water	0.00	0.00	0.00	1,307.70
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	318.77	144.54	838.95	2,656.26
Anticipated crop nutrient removal	224.00	42.00	184.80	0.00
Actual crop nutrient removal	228.89	33.33	195.56	1,020.70
Nutrient balance	89.88	111.20	643.40	1,635.56
Applied to removed ratio	1.39	4.34	4.29	2.60

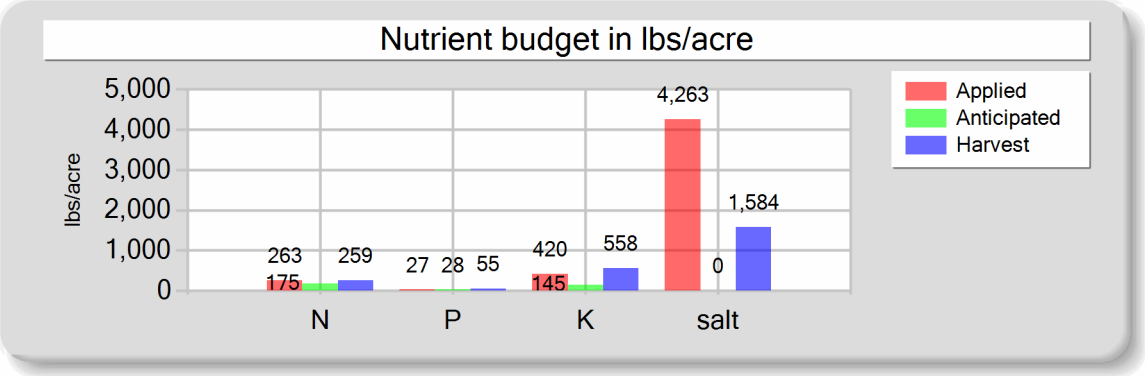
Fresh water applied	
27,297,000.00	*gallons*
1,005.26	*acre-inches*
37.23	*inches/acre*

Process wastewater applied	
4,320,000.00	*gallons*
159.09	*acre-inches*
5.89	*inches/acre*

Total harvests for the crop	
1	*harvests*

Dairy 3 - 11/02/2022: Oats, silage-soft dough

Field name: Dairy 3 Crop: Oats, silage-soft dough Plant date: 11/02/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	255.97	27.23	419.74	3,908.54
Fresh water	0.00	0.00	0.00	354.61
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	262.97	27.23	419.74	4,263.15
Anticipated crop nutrient removal	175.00	28.00	145.25	0.00
Actual crop nutrient removal	259.20	55.38	557.54	1,584.00
Nutrient balance	3.77	-28.15	-137.80	2,679.15
Applied to removed ratio	1.01	0.49	0.75	2.69

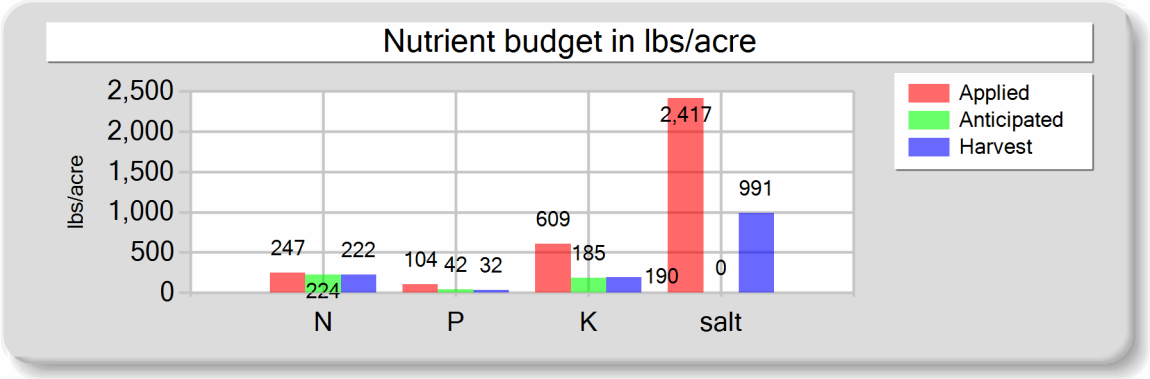
Fresh water applied	
10,692,000.00	*gallons*
393.75	*acre-inches*
10.10	*inches/acre*

Process wastewater applied	
3,960,000.00	*gallons*
145.83	*acre-inches*
3.74	*inches/acre*

Total harvests for the crop	
1	*harvests*

Dairy 3 - 06/01/2023: Corn, silage

Field name: Dairy 3 Crop: Corn, silage Plant date: 06/01/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	239.80	104.26	609.31	1,153.92
Fresh water	0.00	0.00	0.00	1,262.63
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	246.80	104.26	609.31	2,416.55
Anticipated crop nutrient removal	224.00	42.00	184.80	0.00
Actual crop nutrient removal	222.27	32.37	189.90	991.18
Nutrient balance	24.53	71.89	419.41	1,425.37
Applied to removed ratio	1.11	3.22	3.21	2.44

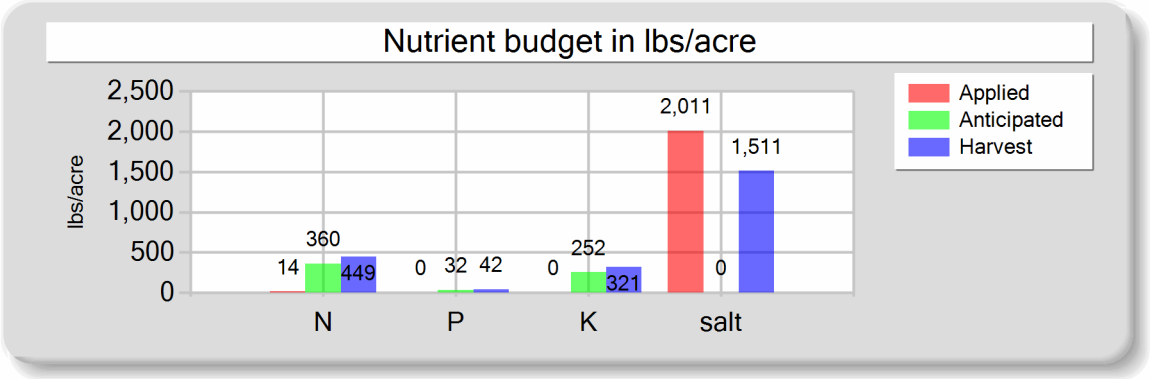
Fresh water applied	
38,070,000.00	*gallons*
1,401.99	*acre-inches*
35.95	*inches/acre*

Process wastewater applied	
4,680,000.00	*gallons*
172.35	*acre-inches*
4.42	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field EV1-3 - 01/01/2023: Alfalfa, hay

Field name: Field EV1-3 Crop: Alfalfa, hay Plant date: 01/01/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	2,010.52
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	14.00	0.00	0.00	2,010.52
Anticipated crop nutrient removal	360.00	32.40	252.00	0.00
Actual crop nutrient removal	448.63	42.42	321.38	1,510.65
Nutrient balance	-434.63	-42.42	-321.38	499.87
Applied to removed ratio	0.03	0.00	0.00	1.33

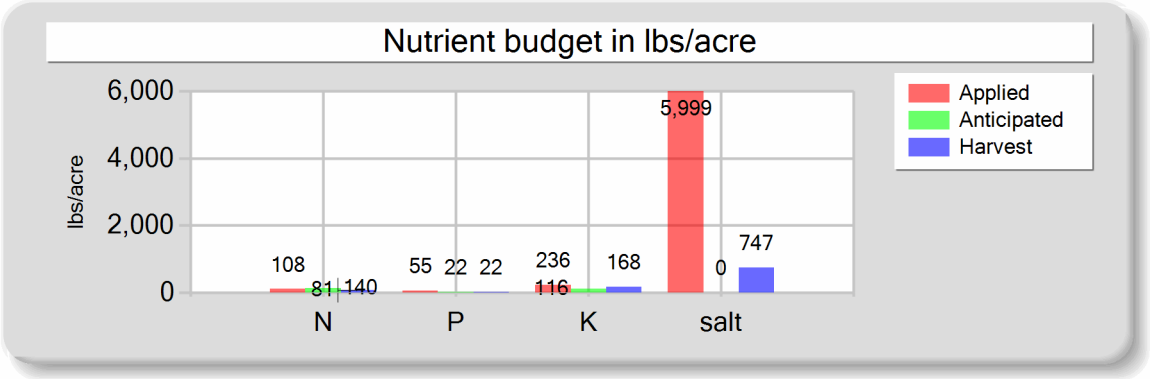
Fresh water applied	
158,544,000.00	*gallons*
5,838.64	*acre-inches*
57.24	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field EV4 - 10/26/2022: Oats, silage-soft dough

Field name: Field EV4 Crop: Oats, silage-soft dough Plant date: 10/26/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	100.75	54.56	235.60	5,859.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	139.70
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	107.75	54.56	235.60	5,998.70
Anticipated crop nutrient removal	140.00	22.40	116.20	0.00
Actual crop nutrient removal	80.72	21.75	167.96	746.75
Nutrient balance	27.03	32.81	67.64	5,251.95
Applied to removed ratio	1.33	2.51	1.40	8.03

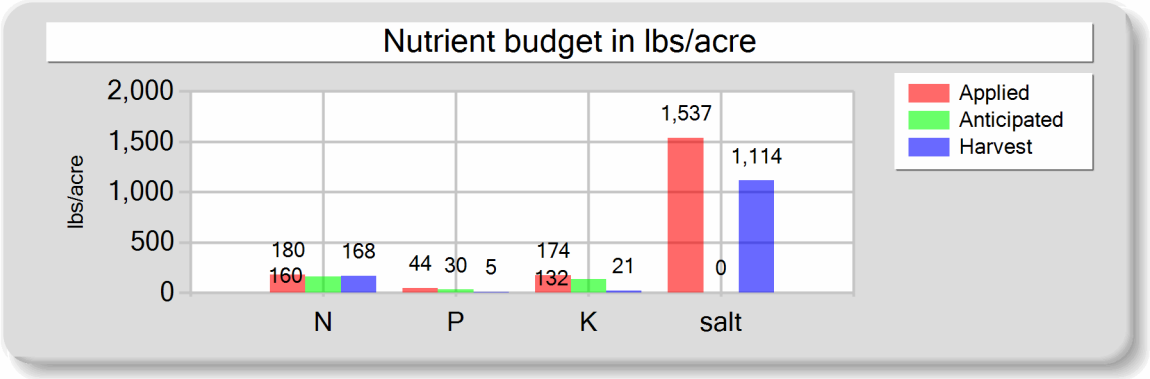
Fresh water applied	
3,888,000.00	*gallons*
143.18	*acre-inches*
3.98	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field EV4 - 06/08/2023: Corn, silage

Field name: Field EV4 Crop: Corn, silage Plant date: 06/08/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	172.87	44.05	173.85	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	1,536.65
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	179.87	44.05	173.85	1,536.65
Anticipated crop nutrient removal	160.00	30.00	132.00	0.00
Actual crop nutrient removal	167.86	5.09	21.44	1,113.81
Nutrient balance	12.01	38.96	152.41	422.84
Applied to removed ratio	1.07	8.66	8.11	1.38

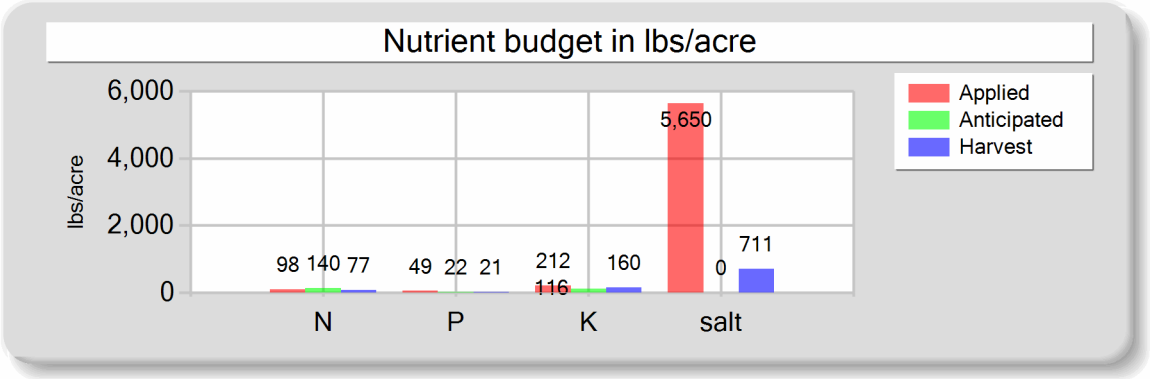
Fresh water applied	
42,768,000.00	*gallons*
1,575.00	*acre-inches*
43.75	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field EV5 - 10/26/2022: Oats, silage-soft dough

Field name: Field EV5 Crop: Oats, silage-soft dough Plant date: 10/26/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	90.68	49.10	212.04	5,273.10
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	377.18
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	97.68	49.10	212.04	5,650.28
Anticipated crop nutrient removal	140.00	22.40	116.20	0.00
Actual crop nutrient removal	76.82	20.70	159.85	710.70
Nutrient balance	20.86	28.40	52.19	4,939.58
Applied to removed ratio	1.27	2.37	1.33	7.95

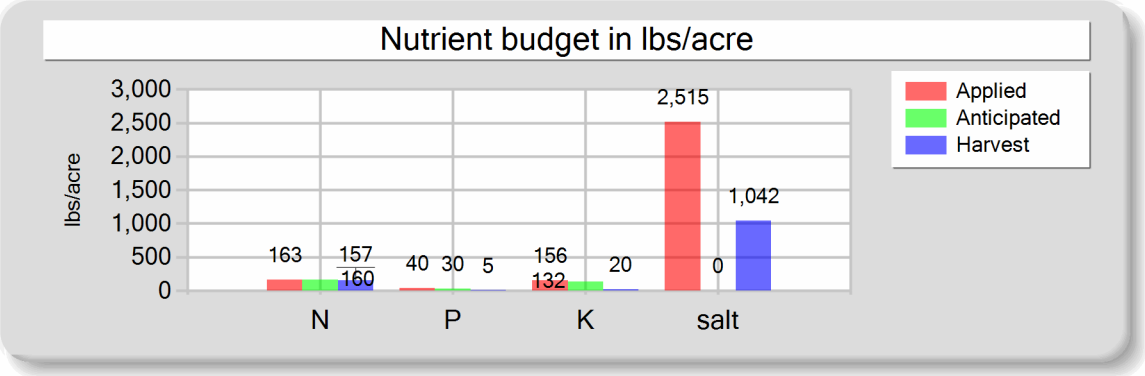
Fresh water applied	
5,832,000.00	*gallons*
214.77	*acre-inches*
10.74	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field EV5 - 06/08/2023: Corn, silage

Field name: Field EV5 Crop: Corn, silage Plant date: 06/08/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	155.58	39.64	156.46	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	2,514.52
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	162.58	39.64	156.46	2,514.52
Anticipated crop nutrient removal	160.00	30.00	132.00	0.00
Actual crop nutrient removal	157.08	4.76	20.06	1,042.28
Nutrient balance	5.50	34.88	136.40	1,472.24
Applied to removed ratio	1.04	8.33	7.80	2.41

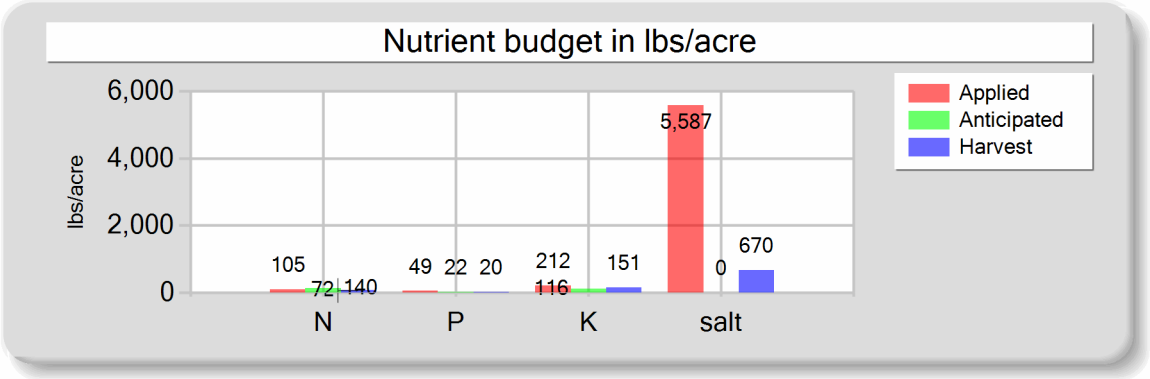
Fresh water applied	
38,880,000.00	*gallons*
1,431.82	*acre-inches*
71.59	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field EV6 - 10/26/2022: Oats, silage-soft dough

Field name: Field EV6 Crop: Oats, silage-soft dough Plant date: 10/26/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	90.68	49.10	212.04	5,273.10
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	314.31
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	104.68	49.10	212.04	5,587.41
Anticipated crop nutrient removal	140.00	22.40	116.20	0.00
Actual crop nutrient removal	72.37	19.50	150.58	669.50
Nutrient balance	32.31	29.60	61.46	4,917.91
Applied to removed ratio	1.45	2.52	1.41	8.35

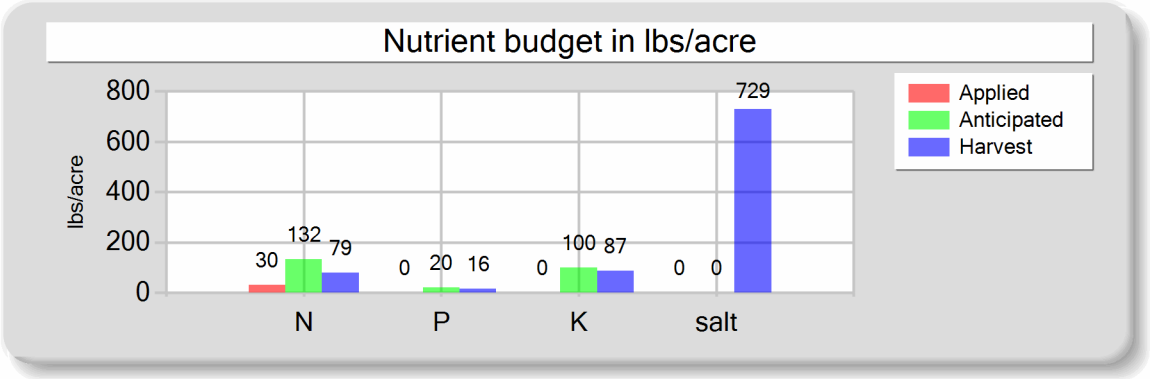
Fresh water applied	
5,832,000.00	*gallons*
214.77	*acre-inches*
8.95	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field P1-2 - 11/09/2022: Wheat Hay

Field name: Field P1-2 Crop: Wheat Hay Plant date: 11/09/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	16.06	0.00	0.00	0.00
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	30.06	0.00	0.00	0.00
Anticipated crop nutrient removal	132.00	20.40	99.60	0.00
Actual crop nutrient removal	78.69	16.28	87.26	729.24
Nutrient balance	-48.63	-16.28	-87.26	-729.24
Applied to removed ratio	0.38	0.00	0.00	0.00

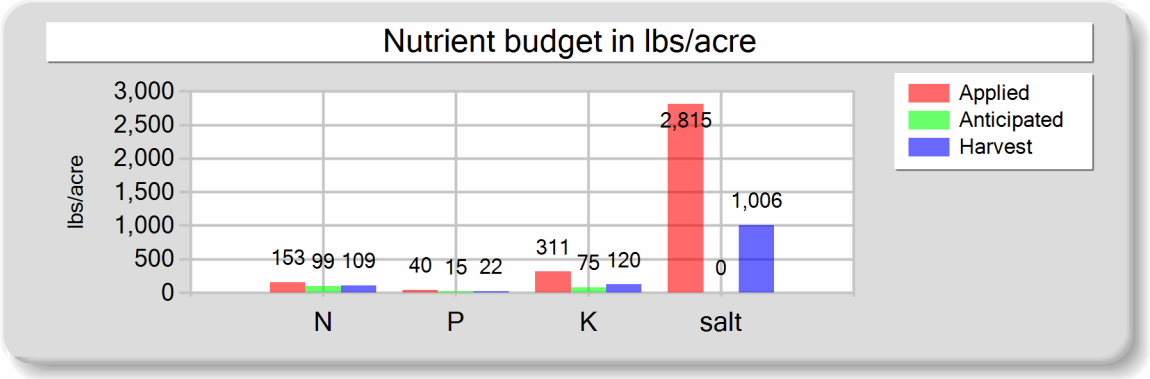
Fresh water applied	
21,384,000.00	*gallons*
787.50	*acre-inches*
11.75	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field P3-4 - 11/10/2022: Wheat Hay

Field name: Field P3-4 Crop: Wheat Hay Plant date: 11/10/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	129.83	40.21	311.33	2,815.20
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	16.52	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	153.35	40.21	311.33	2,815.20
Anticipated crop nutrient removal	99.00	15.30	74.70	0.00
Actual crop nutrient removal	108.54	22.46	120.36	1,005.89
Nutrient balance	44.81	17.75	190.97	1,809.31
Applied to removed ratio	1.41	1.79	2.59	2.80

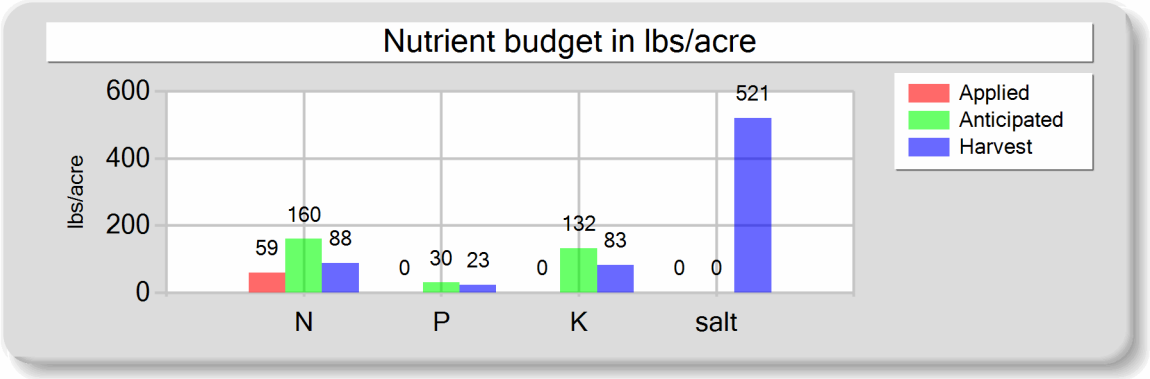
Fresh water applied	
24,624,000.00	*gallons*
906.82	*acre-inches*
12.09	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field P3-4 - 07/02/2023: Corn, silage

Field name: Field P3-4 Crop: Corn, silage Plant date: 07/02/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	52.39	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	59.39	0.00	0.00	0.00
Anticipated crop nutrient removal	160.00	30.00	132.00	0.00
Actual crop nutrient removal	88.37	22.98	83.17	520.93
Nutrient balance	-28.98	-22.98	-83.17	-520.93
Applied to removed ratio	0.67	0.00	0.00	0.00

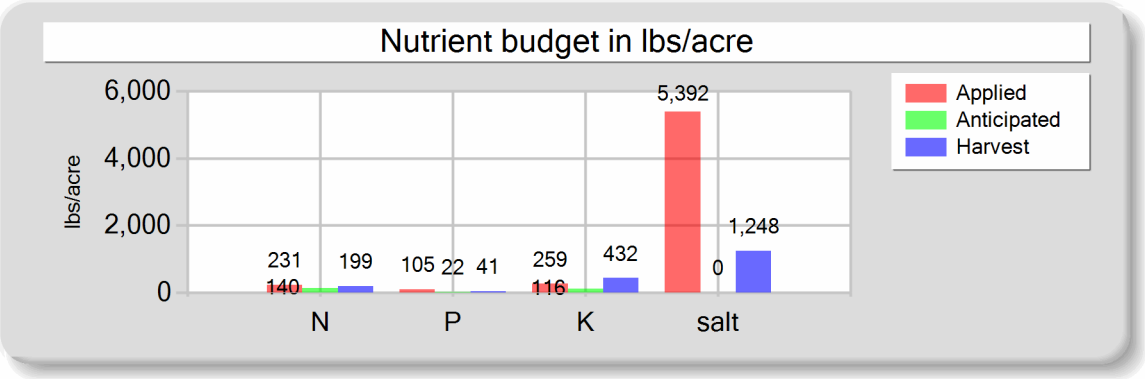
Fresh water applied	
78,084,000.00	*gallons*
2,875.57	*acre-inches*
38.34	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field R-1 - 10/28/2022: Oats, silage-soft dough

Field name: Field R-1 Crop: Oats, silage-soft dough Plant date: 10/28/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	158.48	104.57	259.39	2,418.74
Fresh water	65.14	0.00	0.00	2,972.88
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	230.62	104.57	259.39	5,391.62
Anticipated crop nutrient removal	140.00	22.40	116.20	0.00
Actual crop nutrient removal	198.55	40.97	431.76	1,248.00
Nutrient balance	32.08	63.60	-172.36	4,143.62
Applied to removed ratio	1.16	2.55	0.60	4.32

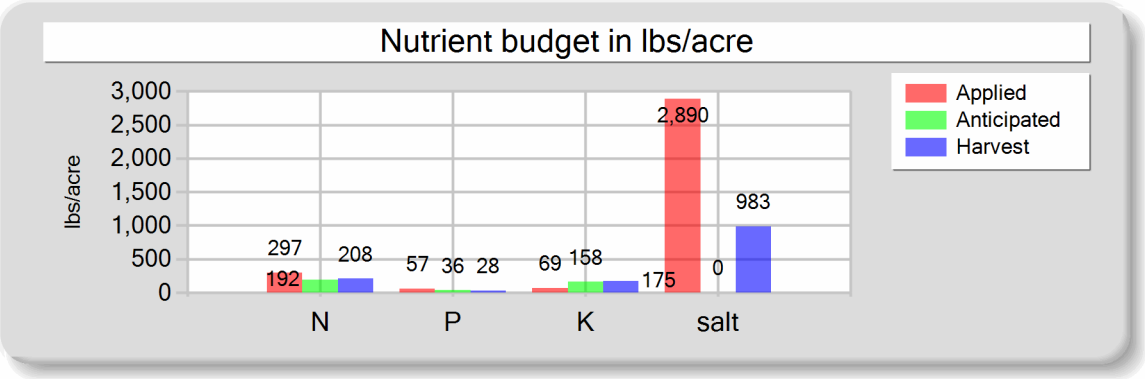
Fresh water applied	
22,464,000.00	*gallons*
827.27	*acre-inches*
8.36	*inches/acre*

Process wastewater applied	
6,240,000.00	*gallons*
229.80	*acre-inches*
2.32	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field R-1 - 06/11/2023: Corn, silage

Field name: Field R-1 Crop: Corn, silage Plant date: 06/11/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	100.10	57.05	68.89	2,890.04
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	190.30	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	297.40	57.05	68.89	2,890.04
Anticipated crop nutrient removal	192.00	36.00	158.40	0.00
Actual crop nutrient removal	207.69	28.07	175.03	982.57
Nutrient balance	89.71	28.98	-106.15	1,907.47
Applied to removed ratio	1.43	2.03	0.39	2.94

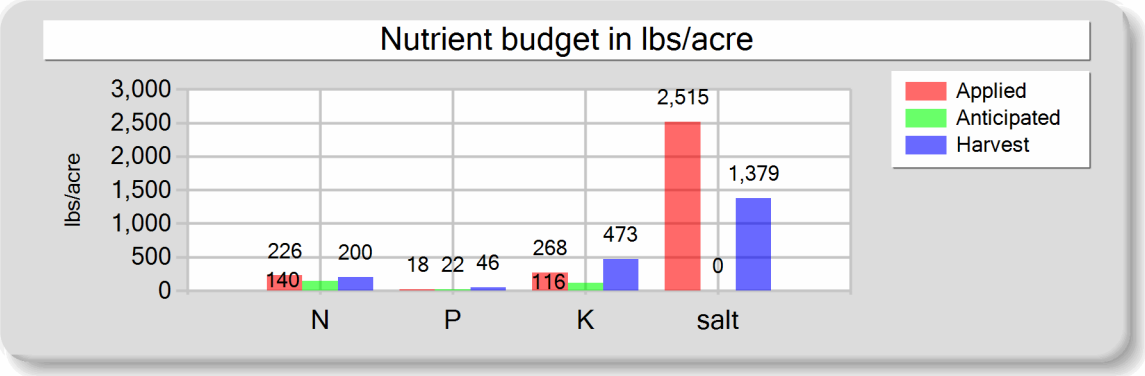
Fresh water applied	
84,240,000.00	*gallons*
3,102.27	*acre-inches*
31.34	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field R-2 - 10/28/2022: Oats, silage-soft dough

Field name: Field R-2 Crop: Oats, silage-soft dough Plant date: 10/28/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	165.23	18.25	267.60	2,514.63
Fresh water	53.68	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	225.91	18.25	267.60	2,514.63
Anticipated crop nutrient removal	140.00	22.40	116.20	0.00
Actual crop nutrient removal	200.33	45.64	473.13	1,379.13
Nutrient balance	25.58	-27.39	-205.54	1,135.49
Applied to removed ratio	1.13	0.40	0.57	1.82

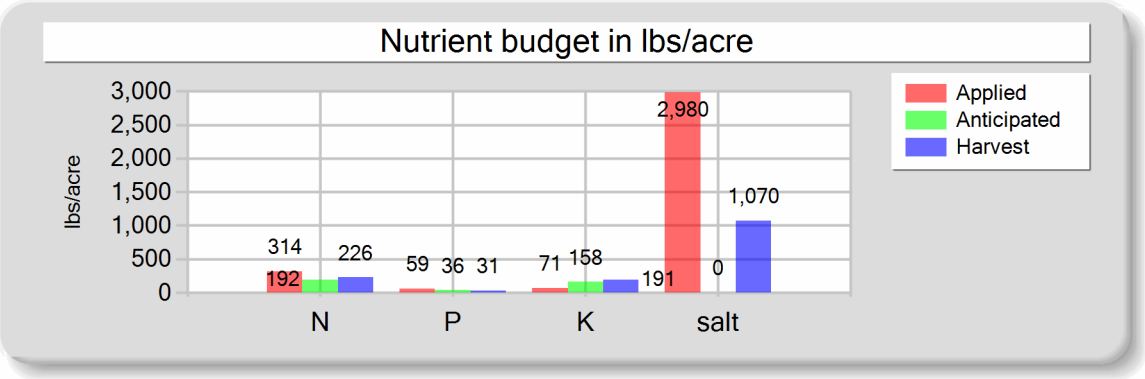
Fresh water applied	
4,320,000.00	*gallons*
159.09	*acre-inches*
8.84	*inches/acre*

Process wastewater applied	
1,200,000.00	*gallons*
44.19	*acre-inches*
2.46	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field R-2 - 06/18/2023: Corn, silage

Field name: Field R-2 Crop: Corn, silage Plant date: 06/18/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	103.23	58.83	71.04	2,980.35
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	203.97	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	314.20	58.83	71.04	2,980.35
Anticipated crop nutrient removal	192.00	36.00	158.40	0.00
Actual crop nutrient removal	226.11	30.56	190.56	1,069.70
Nutrient balance	88.08	28.27	-119.52	1,910.65
Applied to removed ratio	1.39	1.93	0.37	2.79

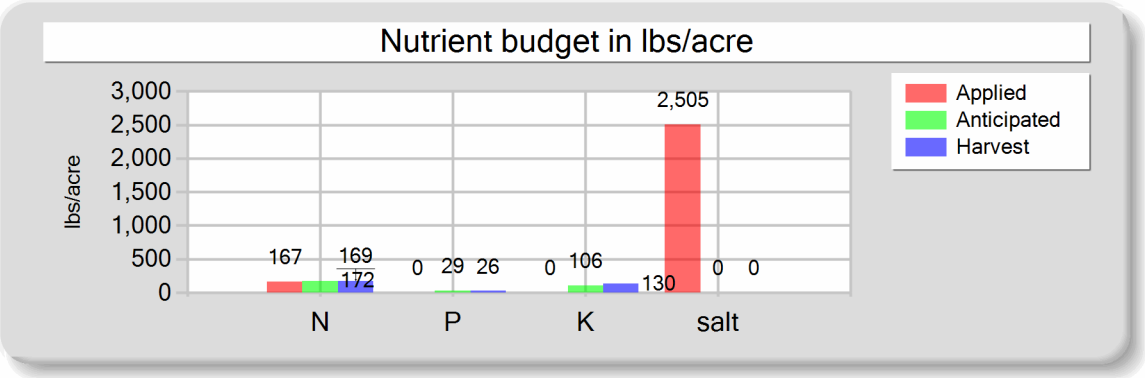
Fresh water applied	
16,416,000.00	*gallons*
604.55	*acre-inches*
33.59	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

Hansen Almonds - 01/01/2023: Almond Field Run

Field name: Hansen Almonds | Crop: Almond Field Run | Plant date: 01/01/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	153.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	2,504.88
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	167.00	0.00	0.00	2,504.88
Anticipated crop nutrient removal	172.00	29.20	106.00	0.00
Actual crop nutrient removal	168.85	25.98	129.89	0.00
Nutrient balance	-1.85	-25.98	-129.89	2,504.88
Applied to removed ratio	0.99	0.00	0.00	0.00

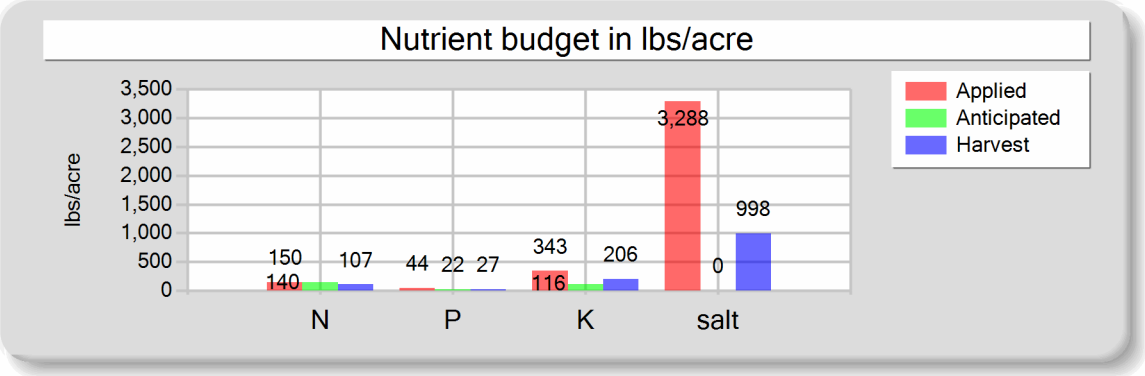
Fresh water applied	
168,480,000.00	*gallons*
6,204.55	*acre-inches*
71.32	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

Hansen Open - 10/30/2022: Oats, silage-soft dough

Field name: Hansen Open | Crop: Oats, silage-soft dough | Plant date: 10/30/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	142.90	44.25	342.66	3,098.49
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	189.77
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	149.90	44.25	342.66	3,288.27
Anticipated crop nutrient removal	140.00	22.40	116.20	0.00
Actual crop nutrient removal	106.75	27.00	205.65	998.40
Nutrient balance	43.15	17.25	137.01	2,289.87
Applied to removed ratio	1.40	1.64	1.67	3.29

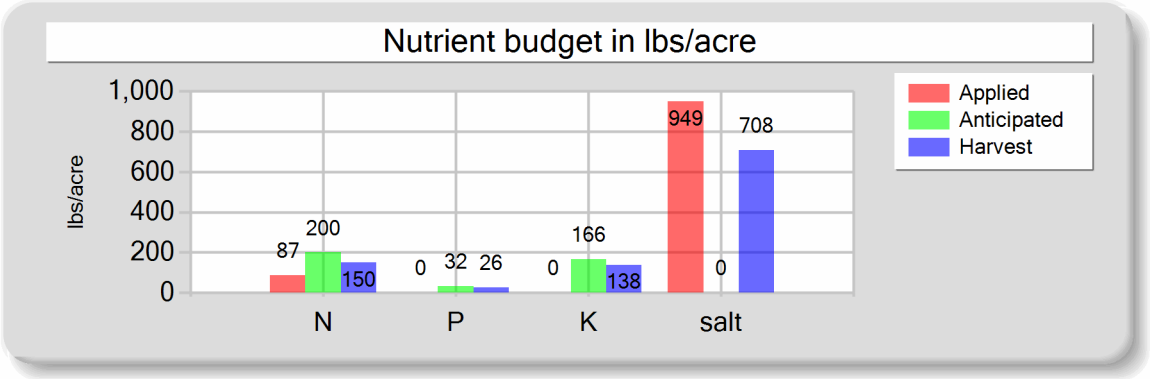
Fresh water applied	
7,776,000.00	*gallons*
286.36	*acre-inches*
5.40	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

Hansen Open - 06/06/2023: Corn, silage

Field name: Hansen Open Crop: Corn, silage Plant date: 06/06/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	80.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	948.87
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	87.00	0.00	0.00	948.87
Anticipated crop nutrient removal	200.00	32.00	166.00	0.00
Actual crop nutrient removal	149.57	25.67	137.59	707.99
Nutrient balance	-62.57	-25.67	-137.59	240.88
Applied to removed ratio	0.58	0.00	0.00	1.34

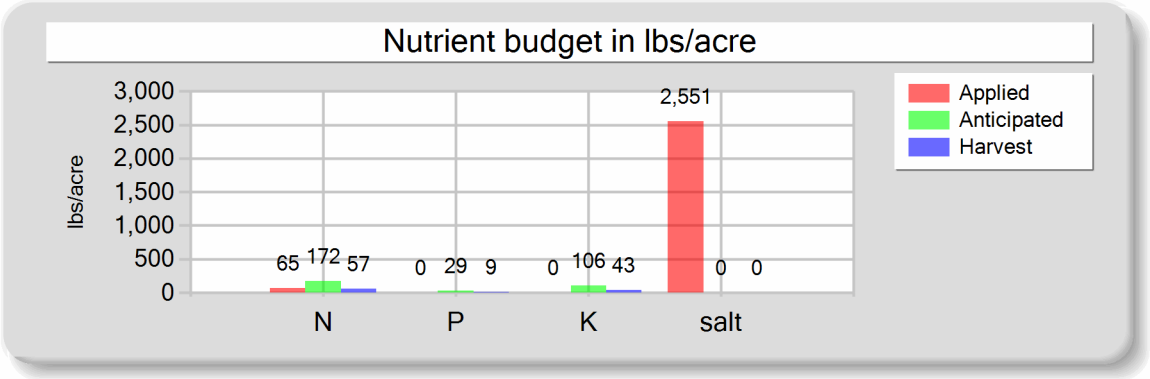
Fresh water applied	
38,880,000.00	*gallons*
1,431.82	*acre-inches*
27.02	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

Ingomar Grade - 01/01/2023: Almond Field Run

Field name: Ingomar Grade | Crop: Almond Field Run | Plant date: 01/01/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	51.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	2,550.96
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	65.00	0.00	0.00	2,550.96
Anticipated crop nutrient removal	172.00	29.20	106.00	0.00
Actual crop nutrient removal	56.52	8.70	43.48	0.00
Nutrient balance	8.48	-8.70	-43.48	2,550.96
Applied to removed ratio	1.15	0.00	0.00	0.00

Fresh water applied	
181,440,000.00	*gallons*
6,681.82	*acre-inches*
72.63	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

NUTRIENT ANALYSES

A. MANURE ANALYSES

Fall SM

Sample and source description: Fall SM

Sample date: 10/07/2022 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 44.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	19,600.00	6,070.00	47,000.00							42.50
DL	100.00	100.00	100.00							0.01

Heifer Facility

Sample and source description: Heifer Facility

Sample date: 10/07/2022 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 44.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	13,000.00	7,040.00	30,400.00							75.60
DL	100.00	100.00	100.00							0.01

Heifer Spring SM

Sample and source description: Heifer Spring SM

Sample date: 05/03/2023 Material type: Separator solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 70.7 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	17,700.00	4,510.00	17,800.00							0.00
DL	100.00	100.00	100.00							0.01

Spring Solid Manure

Sample and source description: Spring Solid Manure

Sample date: 05/03/2023 Material type: Separator solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 66.7 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	18,600.00	10,600.00	12,800.00							53.70
DL	100.00	100.00	100.00							0.01

Fall SM

Sample and source description: Fall SM

Sample date: 10/05/2023 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 47.1 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	7,800.00	5,830.00	34,200.00							86.80
DL	100.00	100.00	100.00							0.01

Heifer Fall SM

Sample and source description: Heifer Fall SM

Sample date: 10/05/2023 Material type: Separator solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 61.7 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	22,300.00	2,850.00	150,000.00							55.70
DL	100.00	100.00	100.00							0.01

B. PROCESS WASTEWATER ANALYSES

4QWW

Sample and source description: 4QWW

Sample date: 12/21/2022 Material type: Process wastewater Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	333.00	251.00	0.00	0.00	69.70	501.00								5,740.00	3,960
DL	0.50	0.50	0.50	0.50	0.50	0.50								1.00	10

1QWW

Sample and source description: 1QWW

Sample date: 03/10/2023 Material type: Process wastewater Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	496.00	333.00	0.00	0.00	60.70	685.00								7,110.00	5,420
DL	0.50	0.50	0.50	0.50	0.50	0.50								1.00	10

2QWW

Sample and source description: 2QWW

Sample date: 05/19/2023 Material type: Process wastewater Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	218.00	160.00	0.00	0.00	119.00	680.00								3,910.00	640
DL	0.50	0.50	0.50	0.50	0.50	0.50								1.00	10

3QWW

Sample and source description: 3QWW

Sample date: 09/27/2023 Material type: Process wastewater Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	311.00	221.00	0.00	0.00	54.50	370.00								5,430.00	2,860
DL	0.50	0.50	0.50	0.50	0.50	0.50								1.00	10

C. FRESH WATER ANALYSES

CCID

Fresh Water

Sample description: Fresh Water

Sample date: 07/25/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	0.00	0.00	0.00								228.00	155
DL	0.50	0.50	0.50								1.00	10

Charleston New Well

Fresh Water

Sample description: Fresh Water

Sample date: 03/10/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	21.40	0.00	0.00								3,940.00	0
DL	0.50	0.50	0.50								10.00	10

Ramos New Well

Fresh Water

Sample description: Fresh Water

Sample date: 10/09/2022 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	34.40	0.00	0.00								3,010.00	1,570
DL	0.50	0.50	0.50								10.00	10

Ramos New Well

Fresh Water

Sample description: Fresh Water

Sample date: 05/24/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	26.80	0.00	0.00								2,560.00	0
DL	0.50	0.50	0.50								10.00	10

SF IW

Fresh Water

Sample description: Fresh Water

Sample date: 02/27/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	6.03	0.00	0.00								1,400.00	0
DL	0.50	0.50	0.50								10.00	10

D. SOIL ANALYSES

No soil analyses entered.

E. PLANT TISSUE ANALYSES

Charleston East - 11/07/2022: Oats, silage-soft dough

Charleston East - 11/07/2022: Oats, silage-soft dough

Forage

Sample and source description: Forage

Sample date: 05/04/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 70.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	5,360.00	880.00	6,820.00		11.10
DL	100.00	100.00	100.00		0.01

Charleston East - 06/17/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 11/03/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 65.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,130.00	600.00	3,680.00		6.34
DL	100.00	100.00	100.00		0.01

Charleston West - 11/07/2022: Wheat Hay

Forage

Sample and source description: Forage

Sample date: 05/04/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 70.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	5,360.00	880.00	6,820.00		11.10
DL	100.00	100.00	100.00		0.01

Charleston West - 06/26/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 11/03/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 65.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,130.00	600.00	3,680.00		6.34
DL	100.00	100.00	100.00		0.01

Dairy 1 - 11/02/2022: Oats, silage-soft dough

Forage

Sample and source description: Forage

Sample date: 05/04/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 70.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	6,610.00	1,270.00	13,500.00		17.40
DL	100.00	100.00	100.00		0.01

Dairy 1 - 06/16/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 11/03/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 65.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,120.00	600.00	3,520.00		5.31
DL	100.00	100.00	100.00		0.01

Dairy 2 - 11/02/2022: Oats, silage-soft dough

Forage

Sample and source description: Forage

Sample date: 05/04/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 70.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	7,190.00	1,350.00	13,800.00		14.00
DL	100.00	100.00	100.00		0.01

Dairy 2 - 06/07/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 11/03/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 65.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,120.00	600.00	3,520.00		5.31
DL	100.00	100.00	100.00		0.01

Dairy 3 - 11/02/2022: Oats, silage-soft dough

Forage

Sample and source description: Forage

Sample date: 05/04/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 70.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	7,020.00	1,500.00	15,100.00		14.30
DL	100.00	100.00	100.00		0.01

Dairy 3 - 06/01/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 11/03/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 65.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,120.00	600.00	3,520.00		5.31
DL	100.00	100.00	100.00		0.01

Field EV1-3 - 01/01/2023: Alfalfa, hay

Alfalfa

Sample and source description: Alfalfa

Sample date: 09/29/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 7.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	27,500.00	2,600.00	19,700.00		10.00
DL	100.00	100.00	100.00		0.01

Field EV4 - 10/26/2022: Oats, silage-soft dough

Forage

Sample and source description: Forage

Sample date: 05/04/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 70.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,340.00	900.00	6,950.00		10.30
DL	100.00	100.00	100.00		0.01

Field EV4 - 06/08/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 11/03/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 57.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,620.00	140.00	590.00		7.23
DL	100.00	100.00	100.00		0.01

Field EV5 - 10/26/2022: Oats, silage-soft dough

Forage

Sample and source description: Forage

Sample date: 05/04/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 70.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,340.00	900.00	6,950.00		10.30
DL	100.00	100.00	100.00		0.01

Field EV5 - 06/08/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 11/03/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 57.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,620.00	140.00	590.00		7.23
DL	100.00	100.00	100.00		0.01

Field EV6 - 10/26/2022: Oats, silage-soft dough

Forage

Sample and source description: Forage

Sample date: 05/04/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 70.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,340.00	900.00	6,950.00		10.30
DL	100.00	100.00	100.00		0.01

Forage

Sample and source description: Forage

Sample date: 11/03/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 57.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,620.00	140.00	590.00		7.23
DL	100.00	100.00	100.00		0.01

Field P1-2 - 11/09/2022: Wheat Hay

Forage

Sample and source description: Forage

Sample date: 09/29/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 6.9 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	28,700.00	2,480.00	17,500.00		11.70
DL	100.00	100.00	100.00		0.01

Field P3-4 - 11/10/2022: Wheat Hay

Field P3-4 - 11/10/2022: Wheat Hay

Wheat Hay

Sample and source description: Wheat Hay

Sample date: 09/29/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 6.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	10,100.00	2,090.00	11,200.00		10.00
DL	100.00	100.00	100.00		0.01

Field P3-4 - 07/02/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 11/03/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 66.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,230.00	840.00	3,040.00		5.60
DL	100.00	100.00	100.00		0.01

Field R-1 - 10/28/2022: Oats, silage-soft dough

Forage

Sample and source description: Forage

Sample date: 05/04/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 70.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	6,300.00	1,300.00	13,700.00		13.20
DL	100.00	100.00	100.00		0.01

Field R-1 - 06/11/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 11/03/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 65.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,070.00	550.00	3,430.00		5.63
DL	100.00	100.00	100.00		0.01

Field R-2 - 10/28/2022: Oats, silage-soft dough

Forage

Sample and source description: Forage

Sample date: 05/04/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 70.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	5,970.00	1,360.00	14,100.00		13.70
DL	100.00	100.00	100.00		0.01

Field R-2 - 06/18/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 11/03/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 65.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,070.00	550.00	3,430.00		5.63
DL	100.00	100.00	100.00		0.01

Hansen Open - 06/06/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 11/03/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 59.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,370.00	750.00	4,020.00		5.07
DL	100.00	100.00	100.00		0.01

F. SUBSURFACE (TILE) DRAINAGE ANALYSES

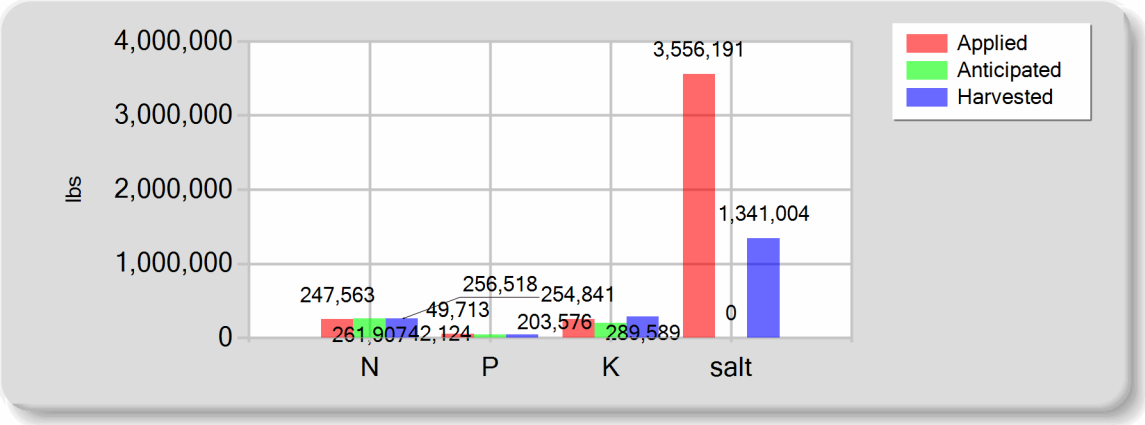
No subsurface (tile) drainage analyses entered.

NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

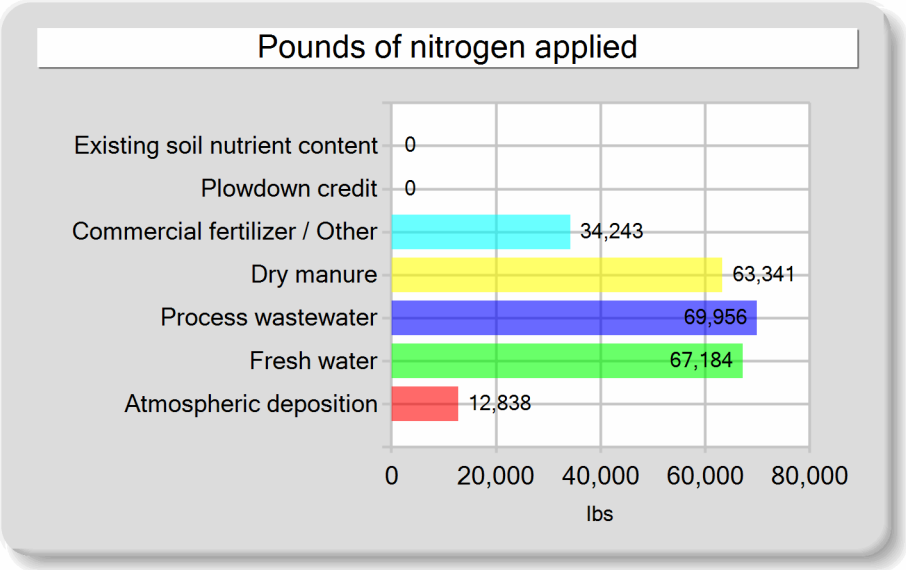
A. SUMMARY OF NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

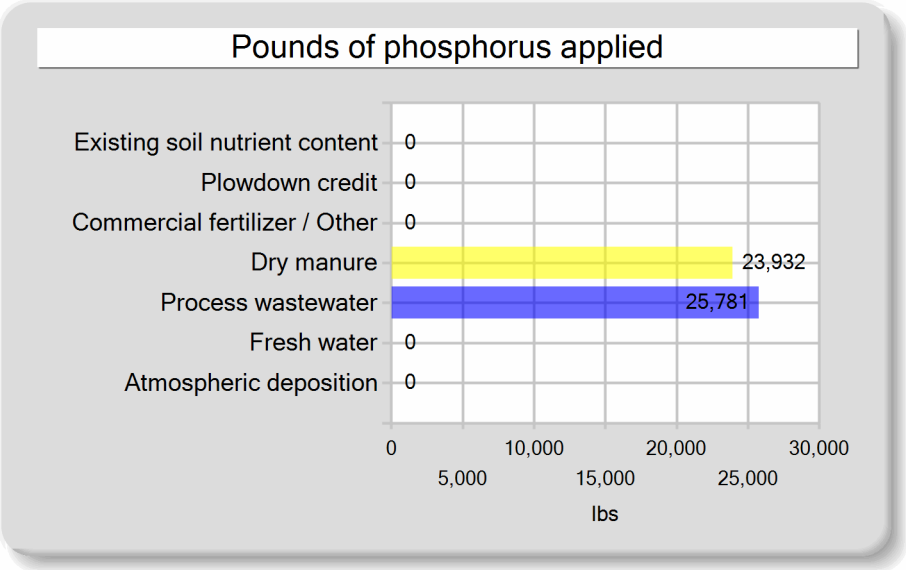
	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	34,243.00	0.00	0.00	0.00
Dry manure	63,341.34	23,931.98	118,318.44	1,533,420.30
Process wastewater	69,956.30	25,781.49	136,522.36	732,043.43
Fresh water	67,183.95	0.00	0.00	1,290,727.12
Atmospheric deposition	12,838.00	0.00	0.00	0.00
Total nutrients applied	247,562.59	49,713.47	254,840.80	3,556,190.84
Anticipated crop nutrient removal	261,907.00	42,123.90	203,576.20	0.00
Actual crop nutrient removal	256,517.58	40,466.86	289,588.98	1,341,004.38
Nutrient balance	-8,954.99	9,246.61	-34,748.18	2,215,186.46
Applied to removed ratio	0.97	1.23	0.88	2.65

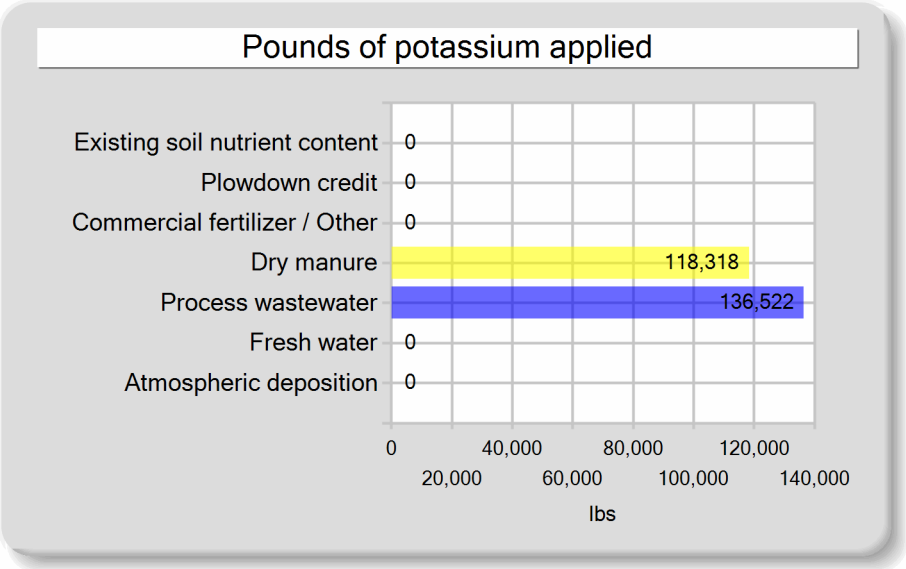
B. POUNDS OF NUTRIENT APPLIED VS. CROP REMOVAL

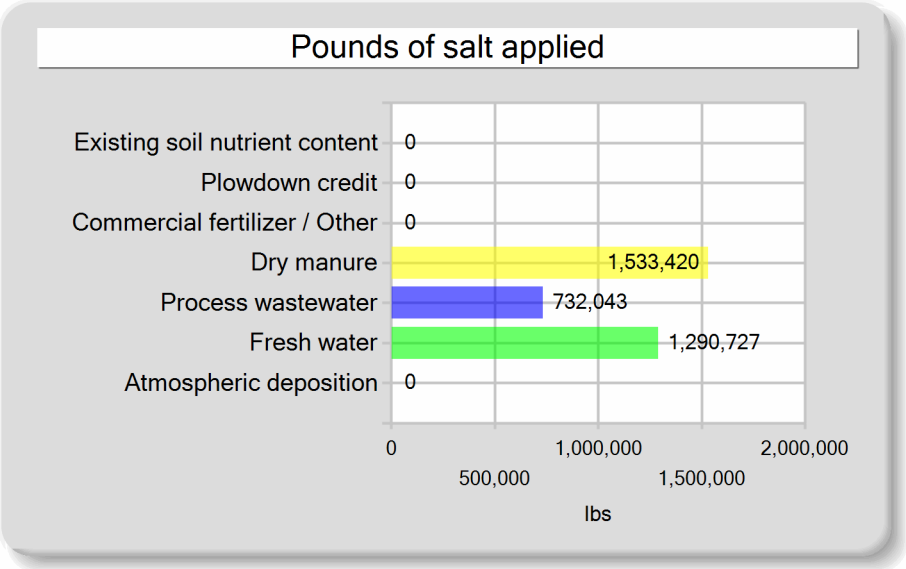


C. POUNDS OF NUTRIENT APPLIED BY MATERIAL TYPE









EXCEPTION REPORTING

A. MANURE, PROCESS WASTEWATER, AND OTHER DAIRY WASTE DISCHARGES

The following is a summary of all manure and process wastewater discharges from the production area to surface water or to land areas (land application areas or otherwise) when not in accordance with the facility's Nutrient Management Plan.

No manure or process wastewater discharges occurred during the reporting period.

B. STORM WATER DISCHARGES

The following is a summary of all storm water discharges from the production area to surface water during the reporting period when not in accordance with the facility 's Nutrient Management Plan.

No stormwater discharges occurred during the reporting period.

C. LAND APPLICATION AREA TO SURFACE WATER DISCHARGES

The following is a summary of all discharges from the land application area to surface water that have occurred during the reporting period when not in accordance with the facility's Nutrient Management Plan.

No land application area to surface water discharges occurred during the reporting period.

NUTRIENT MANAGEMENT PLAN AND EXPORT AGREEMENT STATEMENTS

A. NUTRIENT MANAGEMENT PLAN STATEMENTS

Was the facility's NMP updated in the reporting period?	No
Was the facility's NMP developed by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes
Was the facility's NMP approved by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes

B. EXPORT AGREEMENT STATEMENT

Are there any written agreements with third parties to receive manure or process wastewater that are new or were revised within the reporting period?	No

ADDITIONAL NOTES

A. NOTES

No notes entered for this annual report.

CERTIFICATION

A. OWNER AND/OR OPERATOR CERTIFICATION

I certify under penalty of law that I have personally examined and am familiar with the information submitted in this document and all attachments and that, based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of fine and imprisonment.

Evaristo Vaz (Jun 22, 2024 08:39 PDT)

SIGNATURE OF OWNER OF FACILITY	SIGNATURE OF OPERATOR OF FACILITY
Evaristo Vaz	SAME AS OWNER
PRINT OR TYPE NAME	PRINT OR TYPE NAME
Jun 22, 2024	
DATE	DATE

ATTACHMENTS

A. REQUIRED ATTACHMENTS

The following lists the required documents that should be attached to the Annual Report when submitted .

Annual Dairy Facility Assessment
Provide an Annual Dairy Facility Assessment (an update to the Preliminary Dairy Facility Assessment in Attachment A) for each reporting period. On the PDFA Final page, click on the ADFA Report button to generate an ADFA report after updating information as needed .

Manure/Process Wastewater Tracking Manifests
Provide copies of all manure/process wastewater tracking manifests for the reporting period, signed by both the owner/operator and the hauler.

Corrective Actions Documents
Provide records documenting any corrective actions taken to correct deficiencies noted as a result of the inspections required in the Monitoring Requirements of the General Order. Deficiencies not corrected in 30 days must be accompanied by an explanation of the factors preventing immediate correction.

Groundwater Monitoring
Dischargers that monitor supply wells or subsurface (tile) drainage systems, or that have monitoring well systems must submit monitoring results as directed in the General Order, Groundwater Reporting Section starting on page MRP-13.

Storm Water Monitoring
Dischargers that are required to monitor storm water more frequently than required in the General Order must submit monitoring results as directed in the General Order, Storm Water Reporting Section on page MRP-14.

Denele Analytical, Inc.

865 South Kilroy Avenue Turlock, CA 95380
Phone (209) 634-9055 - Fax (209) 634-9057

www.denelelabs.com

40 N. East Street, Woodland, CA 95776
Phone (530) 666-9056 - Fax (530) 666-9058

Compliance Analysis Report

Customer
Everett Vaz Dairy
21876 W. Sunset Ave.
Los Banos, CA 93635

Grower

Consultant:
F & R Ag Services
2857 Geer Rd. Ste A
Turlock, CA 95382

PURCHASE ORDER:
RECEIVED DATE: 12/28/23 2:00 PM
SUBMITTED BY:
ANALYZED DATE: 1/2/24

Irrigation/ Ground Water (H1)
SOURCE: Well Water

NOTES: Temperature above qualifying at 19.6

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
SF Well	T3362091A	Electrical Conductivity (EC)	EPA 120.1 and	6.03	mmhos/cm
		Nitrate Nitrogen (NO3-N)	EPA 353.2	1.40	mg/L
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Soluble Salts (SALT-SOL)	Calculation	3860	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	RPD %
Ammonia Nitrogen (NH3-N)	91.0			
Electrical Conductivity (EC)	95.4			0.677
Nitrate Nitrogen (NO3-N)	105	99.5 (T3362016J)	98.2 (T3362016J)	1.25

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ELAP Certificate No.: 2714

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Grower

Consultant:
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2857 Geer Rd. Ste A
Turlock, CA 95382

PURCHASE ORDER:
RECEIVED DATE: 12/28/23 2:00 PM
SUBMITTED BY:
ANALYZED DATE: 1/2/24

Irrigation/ Ground Water (H1)
SOURCE: Well Water

NOTES: Temperature above qualifying at 20.8

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
Dairy Dom #2	T3362091B	Electrical Conductivity (EC)	EPA 120.1 and	1.71	mmhos/cm
		Nitrate Nitrogen (NO3-N)	EPA 353.2	18.0	mg/L
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Soluble Salts (SALT-SOL)	Calculation	1100	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	RPD %
Ammonia Nitrogen (NH3-N)	91.0			
Electrical Conductivity (EC)	95.4			0.677
Nitrate Nitrogen (NO3-N)	105	99.5 (T3362016J)	98.2 (T3362016J)	1.25

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Compliance Analysis Report

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Grower

Consultant:
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Turlock, CA 95382

PURCHASE ORDER:
RECEIVED DATE: 12/28/23 2:00 PM
SUBMITTED BY:
ANALYZED DATE: 1/2/24

Irrigation/ Ground Water (H1)
SOURCE: Irrigation Water

NOTES: Temperature above qualifying at 20.0

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
Sunset Ag	T3362091C	Electrical Conductivity (EC)	EPA 120.1 and	2.51	mmhos/cm
		Nitrate Nitrogen (NO3-N)	EPA 353.2	34.2	mg/L
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Soluble Salts (SALT-SOL)	Calculation	1610	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	RPD %
Ammonia Nitrogen (NH3-N)	91.0			
Electrical Conductivity (EC)	95.4			0.677
Nitrate Nitrogen (NO3-N)	105	99.5 (T3362016J)	98.2 (T3362016J)	1.25

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Turlock, CA 95382

PURCHASE ORDER:
RECEIVED DATE: 12/28/23 2:00 PM
SUBMITTED BY:
ANALYZED DATE: 1/2/24

Irrigation/ Ground Water (H1)
SOURCE: Well Water

NOTES: Temperature above qualifying at 18.2

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
Ingomar Dom	T3362091D	Electrical Conductivity (EC)	EPA 120.1 and	1.39	mmhos/cm
		Nitrate Nitrogen (NO3-N)	EPA 353.2	4.84	mg/L
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Soluble Salts (SALT-SOL)	Calculation	890	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	RPD %
Ammonia Nitrogen (NH3-N)	91.0			
Electrical Conductivity (EC)	95.4			0.677
Nitrate Nitrogen (NO3-N)	105	99.5 (T3362016J)	98.2 (T3362016J)	1.25

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Grower

Consultant:
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Turlock, CA 95382

PURCHASE ORDER:
RECEIVED DATE: 12/28/23 2:00 PM
SUBMITTED BY:
ANALYZED DATE: 1/2/24

Irrigation/ Ground Water (H1)
SOURCE: Irrigation Water

NOTES: Temperature above qualifying at 20.2

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
CH Old Ag	T3362091E	Electrical Conductivity (EC)	EPA 120.1 and	3.39	mmhos/cm
		Nitrate Nitrogen (NO3-N)	EPA 353.2	24.2	mg/L
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Soluble Salts (SALT-SOL)	Calculation	2170	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	RPD %
Ammonia Nitrogen (NH3-N)	91.0			
Electrical Conductivity (EC)	95.4			0.677
Nitrate Nitrogen (NO3-N)	105	99.5 (T3362016J)	98.2 (T3362016J)	1.25

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Consultant:
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2857 Geer Rd. Ste A
Turlock, CA 95382

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SUBMITTED BY:
ANALYZED DATE: 1/2/24

Irrigation/ Ground Water (H1)
SOURCE: Well Water

NOTES: Temperature above qualifying at 18.8

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
Volta Dom	T3362091F	Electrical Conductivity (EC)	EPA 120.1 and	1.48	mmhos/cm
		Nitrate Nitrogen (NO3-N)	EPA 353.2	10.4	mg/L
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Soluble Salts (SALT-SOL)	Calculation	945	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	RPD %
Ammonia Nitrogen (NH3-N)	91.0			
Electrical Conductivity (EC)	95.4			0.677
Nitrate Nitrogen (NO3-N)	105	99.5 (T3362016J)	98.2 (T3362016J)	1.25

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Grower

Consultant:
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2857 Geer Rd. Ste A
Turlock, CA 95382

PURCHASE ORDER:
RECEIVED DATE: 12/28/23 2:00 PM
SUBMITTED BY:
ANALYZED DATE: 1/2/24

Irrigation/ Ground Water (H1)
SOURCE: Irrigation Water

NOTES: Temperature above qualifying at 19.4

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
Ramos Ag #1	T3362091G	Electrical Conductivity (EC)	EPA 120.1 and	2.56	mmhos/cm
		Nitrate Nitrogen (NO3-N)	EPA 353.2	26.8	mg/L
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Soluble Salts (SALT-SOL)	Calculation	1640	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	RPD %
Ammonia Nitrogen (NH3-N)	91.0			
Electrical Conductivity (EC)	95.4			0.677
Nitrate Nitrogen (NO3-N)	105	99.5 (T3362016J)	98.2 (T3362016J)	1.25

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Turlock, CA 95382

PURCHASE ORDER:
RECEIVED DATE: 12/28/23 2:00 PM
SUBMITTED BY:
ANALYZED DATE: 1/2/24

Irrigation/ Ground Water (H1)
SOURCE: Irrigation Water

NOTES: Temperature above qualifying at 19.8

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
Ramos Ag #2	T3362091H	Electrical Conductivity (EC)	EPA 120.1 and	4.00	mmhos/cm
		Nitrate Nitrogen (NO3-N)	EPA 353.2	45.1	mg/L
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Soluble Salts (SALT-SOL)	Calculation	2560	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	RPD %
Ammonia Nitrogen (NH3-N)	91.0			
Electrical Conductivity (EC)	95.4			0.677
Nitrate Nitrogen (NO3-N)	105	99.5 (T3362016J)	98.2 (T3362016J)	1.25

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Grower

Consultant:
F & R Ag Services
2857 Geer Rd. Ste A
Turlock, CA 95382

PURCHASE ORDER:
RECEIVED DATE: 12/28/23 2:00 PM
SUBMITTED BY:
ANALYZED DATE: 1/2/24

Irrigation/ Ground Water (H1)
SOURCE: Well Water

NOTES: Temperature above qualifying at 18.6

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
Dairy Dom #1	T3362091I	Electrical Conductivity (EC)	EPA 120.1 and	1.74	mmhos/cm
		Nitrate Nitrogen (NO3-N)	EPA 353.2	17.2	mg/L
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Soluble Salts (SALT-SOL)	Calculation	1110	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	RPD %
Ammonia Nitrogen (NH3-N)	91.0			
Electrical Conductivity (EC)	97.4			0.230
Nitrate Nitrogen (NO3-N)	105	99.5 (T3362016J)	98.2 (T3362016J)	1.25

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RECEIVED DATE: 12/28/23 2:00 PM
SUBMITTED BY:
ANALYZED DATE: 1/2/24

Irrigation/ Ground Water (H1)
SOURCE: Irrigation Water

NOTES: Temperature above qualifying at 18.0

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
CH New Ag	T3362091J	Electrical Conductivity (EC)	EPA 120.1 and	3.94	mmhos/cm
		Nitrate Nitrogen (NO3-N)	EPA 353.2	21.4	mg/L
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Soluble Salts (SALT-SOL)	Calculation	2520	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	RPD %
Ammonia Nitrogen (NH3-N)	91.0			
Electrical Conductivity (EC)	97.4			0.230
Nitrate Nitrogen (NO3-N)	105	99.5 (T3362016J)	98.2 (T3362016J)	1.25

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DENELE ANALYTICAL, INC.

865 S. KILROY
TURLOCK, CA 95380

PH# 209-634-9055 FAX# 209-634-9057

DAIRY COMPLIANCE

Client: Everett Vaz Dairy
Mailing address: 2187 W. Sunset Ave
(not facility address) Los Banos, CA
Phone#:

Authorized Copy Released To: FHR

Sampled By:
Project ID:

ANALYSIS TO BE COMPLETED:

IRRIGATION/GROUND WATER
- H1 EC*, NO3-N, NH4-N*
- H6 EC, TN, TDS
- H7 Combination of H1 & H6
- H14 EC, NO3-N, NH4-N, Ca, Mg, Na, Bicarb Carb, SO4, Cl, TDS

PROCESS WASTE WATER
- H3 EC*, NH4-N, TKN, TP, K, TDS
- H4 EC*, NO3-N, NH4-N, TKN, TP, K, TDS
- H5 Ca, Mg, Na, Bicarb, Carb, SO4, Cl
- H8 Combination of H3 &H5
- H9 Combination of H4 &H5

TILE DRAIN
- H2 EC*, NO3-N, TP, NH4-N*, TDS

** Required Field Measurement*

___ (Check here if performed)

DISCHARGE WATER
- E1 EC*, Temp*, pH*, NO3-N, NH4-N, TKN, TP, K, TDS, Dissolved Oxygen*, BOD 5, TSS, Total & Fecal Coliforms
- E2 EC,* Temp*, pH*, NO3-N, NH4-N*, Turbidity, TP, Dissolved Oxygen*, Total & Fecal Coliforms
- E3 EC*, Temp*, pH*, NO3-N, NH4-N*, TP, Total & Fecal Coliforms
- E4 EC*, Temp*, pH*, NO3-N, NH4-N*, Turbidity, TP, Total & Fecal Coliform

MANURE
- M1 % Moisture
- M3 TN TP, K, % Moisture
- M4 Ca, Mg, Na, Bicarb, Carb, SO4, Cl, Ash
- M7 Combination of M3 & M4

PLANT TISSUE
- F18 % Moisture, Ash, TN, P, K, DM and as received
- F18+ % Moisture, Ash, TN, P, K, DM and as received (Package also includes NIR DCAD)
- F19+ % Moisture, TN (Package includes complete plant tissue analysis)

SOIL
- K3 Soluble Phosphorus
- K4 1ft NO3, Organic Matter
- K5 Pre-Plant (2ft NO3), Fall (1-2ft NO3), Fall (2-3ft NO3)
- K6 0-1ft EC, NO3, Soluble P, K, Organic Matter
- K7 Combination of K3 & K4
- K8 Combination of K3 & K5

	Date Collected	Time Collected	Sample ID	Description	Temp	Well Purge	Analysis: H1	H3	M3	F18	OTHER
1	12/18		SF Well		19.6		✓				
2			Dairy Dom #2		20.8		✓				
3			Sunset Ag		20.0		✓				
4			Ingomar Dom		18.2		✓				
5			CH Old Ag		20.2		✓				
6			Volta Dom		18.8		✓				
7			Ramos Ag #1		19.4		✓				
8			Ramos Ag #2		19.8		✓				
9			Dairy Dom #1		18.6		✓				
10			CH New Ag		19.0		✓				

CHAIN OF CUSTODY RECORDING

	Signature	Company	Received Date & Time	Relinquished Date & Time	Logged in By:	LABORATORY NUMBER (LABORATORY USE ONLY)
1st						
2nd		DENELE ANALYTICAL, INC.	12:28:23 2100			T3362091

Required to be submitted to any agency? (YES / NO)

Vaz Dairy 2023 Annual Report Final

Final Audit Report 2024-06-22

Created:	2024-06-21
By:	Joe Ramos (jramos@fragservices.com)
Status:	Signed
Transaction ID:	CBJCHBCAABAAb-ZPNsZAxZgKlEda-p8Y9mCuDhHLonda

"Vaz Dairy 2023 Annual Report Final" History

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Signer vazdairy@gmail.com entered name at signing as Evaristo Vaz
2024-06-22 - 3:39:34 PM GMT

Document e-signed by Evaristo Vaz (vazdairy@gmail.com)
Signature Date: 2024-06-22 - 3:39:36 PM GMT - Time Source: server

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